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Advanced Persistent Threat detection through Sequential Analysis of Network patterns with Graph based Learning approach

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Advanced Persistent Threat (APT) attack pose significant challenges due to their sophisticated, multi-stage attack patterns, particularly in the rapidly proliferating Industrial Internet of Things (IIoT) environment. The complex nature of APT disrupts the functions of the operational system and causes severe damage to the network environment. Several Machine Learning and Deep Learning models are deployed to detect the APT attack patterns however, they often fail to capture the nuanced evolution of APT attacks and their intricate attack patterns. Considering this, a novel Sequential Analysis of Network patterns with Graph-based Learning approach (SANGL) is proposed for robust APT attack detection and mitigation in IIoT environments. The proposed approach integrates three phases: (i) PyPCA method to extract the critical APT attack patterns. (ii) Motif hypergraph method is deployed to cluster higher-order relationships between APT attack stages using motif trinity process, pretrained using motif augmentation process and top-k motifs are selected using attention mechanism. (iii) Multi-objective Gradient Modulator (MoGraM) based Optimized Graph Convolution Network for enhanced detection of evolving APT behaviours and accurate modelling of APT attack stage-wise transitions. The supremacy of the SANGL approach is validated using various datasets such as CICAPT IIoT, DAPT20, Unraveled, and Edge IIoT and compared with state-of-the-art models in terms of Accuracy, Precision, Detection Rate, F1-score, FPR, Attention Divergence Score, Stage Contrastiveness score, and Stage Transition accuracy. The results demonstrate that the proposed approach significantly improves the APT detection process and enhances the cyber defence mechanism.

Keywords Advanced Persistent Threat, Pyramidal Principal Component Analysis, Motif Hypergraph, Motif Trinity, Optimized GCN, APT attack stage-wise Transition analysis

The Industrial Internet of Things (IIoT) has modernised the industry automation process by enabling seamless data transfer and device interconnectivity through the convergence of Information Technology (IT) and Operational Technology (OT). This interconnectivity enlarged the attack surface, and IIoT systems were hence vulnerable to cyber-attacks, specifically to APT attacks^{1–3}. It poses a unique and formidable challenge, since traditional cyber-attacks like DDoS, DoS, SQL injection, and phishing differ drastically from APT attacks. In general, APT attacks take advantage of hardware, software, or communication protocol vulnerabilities to stealthily gain access to critical infrastructures undetected for months or even years. Recent reports^{4–7} demonstrate that APT attack groups and financially motivated actors are increasingly launching highly sophisticated attacks against supply-chain compromises, zero-day exploits, and cross-platform ransomware affecting critical systems such as ERP software and industrial control servers. Precisely, Sandworm, APT41, and UAC-0063 ponder the invasion of attack vectors in IIoT environments. The diversity of IIoT architectures, especially the presence of legacy OT systems, makes the identification and mitigation of the threat exceptionally complex and challenging.

Despite considerable research efforts, the detection of APT attacks in IIoT environments remains a significant challenge. The multi-stage attack nature of APT attacks, diverse forms of IIoT data, and constrained resources of IIoT devices result in infeasible monitoring and threat correlation^{8,9}. Massive amounts of data from sensor-related protocols, flag sequences, and temporal information are generated from IIoT environments. The

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diversified landscape of IIoT data represents severe complications for detection approaches. Traditional rule-based systems are overwhelmed and hence proven to be inefficient to address the underlying problem complexity. Other approaches, such as ML and DL methods, also have their own limitations, including imbalanced data sets, inherent lack of contextual understanding, and poor capturing of complex patterns within IIoT data, thus resulting in poor performance^{10–16}. In conventional models, normal industrial operations were misclassified as malicious, resulting in high false-positive rates. Similarly, conventional models degraded the efficiency of the entire detection process. However, ensemble learning, meta-learning, and semantic analysis-based approaches are promising in the detection process but have specific challenges in the detection of particular APT stages and adaptation to different IIoT environments^{17–21}.

To address these limitations, a new Sequential Analysis of Network patterns with Graph-based Learning approach has been proposed, namely SANGL. It consists of a Pyramidal Principal Component Analysis model for an efficient feature engineering process, a motif-based graph-clustering model for enhanced identification of the interrelation between the stages of APT attacks, and an optimized Graph Convolutional Network to detect the transition between the attack phases. The proposed SANGL approach integrates the sequential and relational graph modelling. Sequential network analysis will capture the temporal behaviour of APT attack patterns, while graph-based learning will capture the higher-order relationships among all the entities and motifs that formed across the stages of the APT attack. The following Research Observations (RO) are discussed to ensure the significance of the SANGL approach over the state-of-the-art APT detection approaches.

- **RO1:** How effectively PyPCA method enhance the identification of critical APT behavioural patterns when compared to conventional feature extraction techniques?
- **RO2:** Does motif hypergraph clustering improve the modelling of higher-order inter-stage relationships in multi-stage APT attacks?
- **RO3:** How does MoGraM influence the convergence stability, adaptability, and detection accuracy of the Graph Convolutional Network (GCN) when handling evolving and obfuscated APT attack patterns?

With these insights into the research observations, the three major contributions of the proposed approach are highlighted below,

- A novel motif-trinity-based graph-clustering model is proposed to capture the higher-order relationships across APT stages.
- An Inter-Stage Contrastive Loss is developed to effectively learn the APT stage-specific semantics and improve the model robustness against noise and obfuscation in IIoT environments.
- MoGraM is implemented to fine-tune the GCN for precise classification and adaptively respond to evolving APT threat patterns.

The remaining sections of the article are structured as follows: Section 2 discusses the related works. Section 3 deliberates the materials and methods. Section 4 elaborates on the working of the proposed approach. Section 5 presents the experimental results, and Section 6 concludes the paper with a discussion of future findings.

Related works

This section reviews existing research on APT detection, highlighting limitations and challenges across various timeframes. Previous works have employed methods such as machine learning and graph-based approaches, but often struggle with complexity, scalability, and adaptability. These limitations underscore the need for more effective APT detection methods. The following subsection elaborates on the previous works based on the Motif hypergraph, GCN and various deep learning methods.

Kołodziej et al.²² proposed a network topology-based motif method for characterising the communication graphs. It provides valuable insights into network topology; however, it exhibits more computation complexity of $O(|V|^3)$ for sparse matrices. Xiuming Zhao et al.²³ proposed a motif attention-based weighting framework for the neural network to capture the complex network structure using multiple feedforward layers with an attention framework. Nevertheless, a lack of analysis of the complex network patterns in the large network data might result in misclassification. Also, the interpretation of traffic patterns is difficult without domain-specific knowledge in larger data. Maidi Liu et al.²⁴ proposed an operational motif-based spectral clustering for seizing the higher-order network patterns. However, this method failed to address computations of multiple operational motifs due to the opaque hyper-graphical structure. In addition, the complex dimensional data creates a motif extraction that is complicated for analysis. Ying wang et al.²⁵ proposed a motif graph-based attention mechanism to learn the complex network patterns and to acquire the prominence of the hierarchical network structures. Nevertheless, these methods often use the normal distribution to assume the frequencies of the motif, which leads to inaccurate statistical significance calculations. Lotito et al.²⁶ proposed a mining method to count the occurrence of the higher-order motifs by reducing the error using the statistical techniques and breaking the chain process of motifs. Yet this model failed to address the limitations such as computational complexity, interpretability and scalability. Quintino Francesco Lotito et al.²⁷ proposed a hypergraph motif-based method to extract the fingerprints of hypergraphs at large-scale network data to provide valuable insights into predicting the local patterns; it also helps for the pairwise communication process. Still, the process works better for motif sizes 3 and 4, but not for larger motif graphs, which leads to computational complexity. Francesco William et al.²⁸ proposed a motif graph-based weighted spectral clustering model for interconnected networks; the clusters are formed based on both the directed and bipartite properties of graphs. The proposed method is implemented on the dataset based on the territories and clustered using the Unicode global language. However, the authors failed to address the computational efficiency.

Cho et al.²⁹ proposed a combined deep learning model that integrates the Bidirectional Long Short-Term (BiLSTM) and Graph Convolutional Network (GCN). APT-related IP flows are extracted from other IP flows and reconstructed into a specific IP flow for detection mechanisms. Analysing the IP flows will lead to a misclassification rate, as most of the IPs mimic the behaviour of benign samples. The proposed model overlooks feature redundancy, making a minimal contribution to the temporal features. Cho et al.³⁰ proposed a model which analyse the behavioural profiles using graph networks. The behaviour of each attempt is found by analysing the EventID from the kernel of the observed system. The authors failed to address the issue, as it focuses only on the end activity and lacks multiscale relations. Nguyen Hoa Cuong et al.³¹ proposed an ensemble-based learning model with a looped correlation layer and an attention layer. It integrates the efficiency of Multi-Layer Perceptron (MLP), Correlation Layer (CR) and Graph Attention Network (GAT). Although it can detect APT attacks from the IP-diffused network dataset, it lacks temporal-level features due to higher model complexity. It has more static-level features that affect the performance of the model. Cho et al.³² proposed a Dynamic Graph Convolutional Neural Network (DGCNN) method that comprises two data mining algorithms, namely Bi-LSTM and an Attention network, for the efficient extraction of behaviour-based features. Though the model achieves higher precision over other models discussed in the article, the authors failed to perform multi-class detection. A comprehensive review of existing literature on APT detection using deep learning models is elucidated in Table 1, which summarises the key findings and methodologies of previous works.

From the above-discussed works [Table 1] and subsections, it is inferred that existing methods for APT stage-wise detection are limited by their lack of interpretability, poor performance, and inability to predict stage transitions accurately. In addition, some of the models failed to address the optimal graph structures, failed to incorporate domain knowledge and progression of APT attack. To address these challenges, a novel SANGL approach is proposed that enhances interpretability, performance, and predictive capabilities for APT attack stage transitions. Specifically, the PyPCA method progressively imports the weights, which extract the APT discriminative features effectively. Motif trinity process enables the extraction of high-order attack patterns, and MoGraM strengthens the process of capturing the critical APT attack patterns. With these insights, the Table 2 presents the detailed research summary table to highlight the supremacy of the proposed approach.

Materials and methods
Pyramidal Principal Component Analysis (PyPCA)

Principal Component Analysis (PCA) is a feature extraction technique, used to enhance the performance of the learning model by focusing more on the optimal data samples⁴². However, it poses various limitations such as linearity, sensitivity to scaling and outliers. To address these limitations, Pyramidal Principal Component Analysis (PyPCA) is proposed, which combines entropy-based, correlation-based, and temporal-based weighting methods to provide a more comprehensive understanding of the large volume of complex data. The covariance matrix is formed, where the Eigen vectors are captured, and an attention mechanism is performed to prioritize the principal components. The following steps present the working of PyPCA,

Step 1 Compute weights for the entropy method $weight_{E_i}$, correlation method $weight_{C_i}$ and temporal method $weight_{T_i}$ Weights.

a. Entropy-based weighting:

$$weight_{E_i} = 1 - f_i/\max(f_i) \tag{1}$$

Authors and Ref.	Methodology used	Merits	Limitations
Srinadh et al ³³ .	Deep residual network and hybrid optimization methods	Robust APT detection, Secure traceability of APT logs	The deployment process is complex, with performance overhead
Najah Kalifah Almazmomi ³⁴	Convolution Neural Network – Long Short Term Memory (CNN-LSTM) and Lime mould algorithm	Scalability, High detection rate	Lack of interpretability and computationally inefficient
Weina Niu et al ³⁵ .	RESNET_LSTM and PARALLEL_LSTM	Effectively captures sequential patterns of APT attacks in network data	Resource-intensive, Difficult in modelling complex behaviours
Animesh et al ³⁶ .	Deep Reinforcement Learning	Dynamically analysed the attribution of APT groups	Overfitting issues, Attribution errors
Fargana J. Abdullayeva ³⁷	Autoencoder and softmax regression	Robust model for anomaly-based detection in a cloud environment	Dataset representativeness, lack of optimisation
Do Xuan et al ³⁸ .	Feature Extraction and Representation Learning Model	Novel Feature extraction method, Efficient learning mechanism	Lack of generalization, Data infeasible
B Tang et al ³⁹ .	Improved CNN	Scalable Effective on large, multi-type APT datasets, scalable	Class imbalance issue, lack of explainability
Yue et al ⁴⁰ .	Attack intent-driven and sequence-based learning approach	Improved detection accuracy, integration of heterogeneous data, and building a provenance graph	Complexity, High false positive rate, Scalability issues
Jung et al ⁴¹ .	Cyber clustering module, clustering analysis module and random forest, Deep Neural Network	Effective APT detection, Precise classification	Complexity, Overfitting issues, Limited adaptability
Du et al ⁴⁵ .	Hypergraph-Empowered Tactic-Specific Traffic Clustering for Atomized APT Detection	Analysis of temporal threat patterns	Lack of stage transition dynamics, lack of explainability

Table 1. Previous works on APT detection using deep learning models.

Article details	Methods Used	Stage Transition	High order relationship	Interpretability	Adaptability	Scalability	Other Limitations
Motif-Based Models (Juszczyszyn et al ²² , Zhao et al ²³ , Liu et al ²⁴ .)	Motif clustering, attention weighting, and spectral analysis	✗	Partial	Moderate	Low	Moderate	High computational cost; weak stage-level learning
Hypergraph Motif Learning (Lotito et al ²⁶ , Francesco et al ²⁷ , William et al ²⁸ .)	Hypergraph motif extraction and spectral clustering	✗	✓	Low	Low	Moderate	Limited to small motifs (3–4 nodes); poor scalability
Hybrid Deep Models (Cho et al ^{29,30,32} , Nguyen et al ³¹ .)	CNN-LSTM, BiLSTM-GCN, GAT-MLP hybrids	Partial	✗	Moderate	Moderate	Moderate	Poor temporal modeling; high feature redundancy
Autoencoder/Reinforcement/Residual Models (Almazmomi ³⁴ , Niu et al ³⁵ , Abdullayeva ³⁷)	Unsupervised feature learning and deep residual inference	✗	✗	Low	Low	High	Lack of interpretability; static learning process
Hypergraph-Based APT Detection (Du et al ⁴⁵ .)	Tactic-specific temporal clustering using hypergraphs	Partial	✓	Moderate	Low	Moderate	Does not address multi-stage attack progression
Proposed SANGL Framework	PyPCA + Motif-Trinity + GraphCL + MoGraM Optimization	✓ Explicit stage transition modeling	✓ Higher-order motif learning	✓ High (attention-driven motif ranking)	✓ Adaptive (multi-objective gradient modulation)	✓ Scalable (motif-level graph representation)	Addresses complexity, interpretability, and temporal adaptability gaps

Table 2. Summary of related works with the proposed SANGL approach.

Equation (1) specifies the working of the entropy-based model, where E_i represents the entropy data. Its weights are considered high when they are low, and when they are high, its weights are considered low. f_i represents the importance of the i^{th} features, $\max(f_i)$ denotes the maximum frequency of the feature. Feature importance values are normalized according to their respective importance.

b. Correlation-based weighting:

$$weight_{Ci} = 1 - \frac{1}{n} \sum_n |corr(f_i, f_n)| \quad (2)$$

Equation (2) mentioned that a redundant feature is weighted low, and it supports independent features to be selected. f_i, f_n refers to the i^{th} feature and n^{th} feature. $\frac{1}{n}$ average correlation values across all the features. The total $weight_{Ci}$ combines the correlation among the features considered.

c. Temporal-based weighting:

$$weight_{Ti} = MI(f_i : ts) \quad (3)$$

The feature is selected based on the mutual information between the feature and the time interval for temporal based weight process Eq. (3). This preserves the quantified details on the feature based on time, with temporal dynamics having more weight. MI refers to the mutual information, ts refers to the timestamp, and T refers to the time values.

Step 2 Hybrid weight calculation using Eq. (4)

$$hybridweights = \omega_E weight_{Ei} + \omega_C weight_{Ci} + \omega_T weight_{Ti} \quad (4)$$

Equation⁴. explicitly explains the combination of the weights obtained from the pyramidal model, where the ω_E, ω_C and ω_T refer to the weights of entropy, correlation and temporal weights.

Where ω represents the hyperparameters of each method. The parameters ω_E, ω_C and ω_T are optimized using the Bayesian optimization process such that the hybrid weights are calculated using Eq. (5).

$$hybridweights = 1i.e \omega_E + \omega_C + \omega_T = 1 \quad (5)$$

The Bayesian optimizer is specifically deployed to tune the projection dimensionality d and component weighting cd , which maximizes the feature discriminability between the benign and the attack stages of APT. The objective function is $\mathcal{L}_{obj} = \max_{d, cd} (F1 - score - \alpha.reconstructionerror)$. The Bayesian optimizer uses Gaussian Process Regression (GPR) and Expected Improvement (EI) to navigate the search space, converging within ~25 iterations to determine optimal feature dimensionality and projection for graph-based learning.

Step 3 All the features are scaled using these weights, $WeightedSamples = Samples * hybridweights$

Step 4 Standardization is performed using $Sd = \text{WeightedSamples} - \mu/\sigma$. μ represents the mean of each feature, and σ represents the standard deviation of each feature. The covariance matrix is computed using the weighted values obtained. $C_M = sd^T \cdot \text{diag}(\text{WeightedSamples}) \cdot sd$

Step 5 Using the above formula $top - \text{keigenvectors}$ are defined, then the, C_M is considered for constructing attention-based PCA. The attention mechanism holds the following value, Eq. (6), Compute

$$Q = C_M * W_q, K = C_M * W_k, V = C_M * W_v \quad (6)$$

W_q refers to the weight matrix used to compute the query representation. W_k represents the key representation and W_v refers to the value representation.

Step 6 The Self-Attention mechanism is used based on the weights aggregated from the pyramidal model Eq. (7)

$\text{Attention}(sd) = P_k \text{WeightedSample}((P_k)^T sd)$ (7). More weights are provided to the temporal-based weighting process for robust model enumeration.

Motif hypergraph-based pretrained and gradient-optimized GCN for APT Detection

A hypergraph $\mathcal{H} = (\mathcal{N}, \epsilon)$ is a graphical structure consisting of $\mathcal{N}$ nodes and ϵ hyperedges [Fig. 1]. It has vertices that join more than one node, and this distinguishes a hypergraph from other graphs^{43–45}. From the following figure, it is observed that $\mathcal{N}1$ to $\mathcal{N}6$ represent nodes interconnected with three hyperedges, 1 to 3. $\mathcal{N}2$ has no further connections; therefore, it is isolated, and no specific information can be derived from this node.

Hypergraphs are useful in capturing the complex relationships in heterogeneous data, representing the complicated structures, but they are computationally inefficient and have scalability issues, along with interpretability challenges. To address this, various methods are used; among them, motif hypergraphs present the robust results over other methods [27 & 28]. However, it failed to address the limitations such as flexibility, in-depth learning process, scalability and interpretability. To address this, a novel motif trinity process is proposed, which employs three techniques: (i) Triangular Motif, (ii) Clique motif and (iii) Bi-partite motif. These techniques are used to capture the nuanced attack patterns of APT with local and global patterns. It is also used to analyse the high-range semantic attack patterns, which are used to enhance the robustness and interpretability of the model. Once the motif hypergraphs are formed, a pretraining process is initiated where a motif-level embedding is identified using a self-supervised Graph Contrastive Learning (GraphCL) mechanism. The process includes data augmentation, embedding generation for GCN, contrastive learning and motif ranking. This approach enhances robustness by learning from multiple augmented views of motif substructures, enabling effective generalization across the diverse structural and behavioral patterns inherent in various attack simulation environments. Augmentation in the motif hypergraph is performed using Motif temporal drift, Motif dropout and subgraph masking process. The following elaborates on the working of the motif augmentation process,

• **Motif temporal drift:** In each motif instance, the time delay Δ_{ts} between the sequential events is considered and recorded. This process is represented as $mtf_a = \{HG_{APTd1}, HG_{APTd2}$

$\dots \dots HG_{APTdn}\}$,

where the number of motif instances is formed using Eq. (8).

$$Mt = \{(HG_{APTd1}, ts_1) \dots \dots (HG_{APTd5}, ts_5 + \Delta_{ts4}), \dots, (HG_{APTdn}, ts_n + \sum_{i=1}^{n-1} \Delta_{tsi})\} \quad (8)$$

• **Motif dropout:** The motif probability is considered in relation to the number of nodes and edges presented in the motif hypergraph. If the probability is considered lower, then the particular motif will be dropped. Considering $HG_{APTd} = \{HG_{APT1}, HG_{APT2}, HG_{APT3}, \dots \dots HG_{APTn}\}$ where $HG_{APTn} \in HG_{APTd}$. Then the random number is selected as $r_{HG_APTd} \sim \mathcal{U}(0,1)$. Motif HG_{APTd} is considered only if $r_{HG_APTd} > \mathcal{P}$ in Eq. (9).

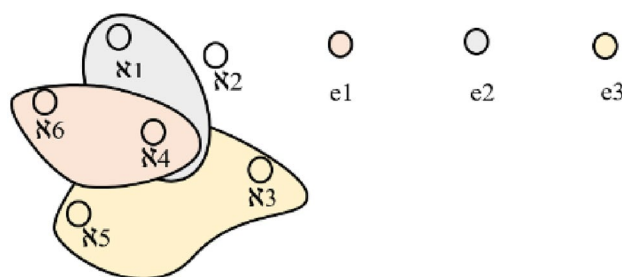


Fig. 1. Structure of Hypergraph.

$$HG'_{APTd} = \{HG_{APT} \in HG_{APTD} | r_{HG_APTd} > \mathcal{P}\} \quad (9)$$

The value of motif dropout probability $\mathcal{P}$ is obtainable with a range from 0.1 to 0.5. If the value of the $\mathcal{P}$ is high ($= 0.5$ or > 0.5), then the motifs are dropped, which simulates the sparse logs. If the value of the $\mathcal{P}$ is low ($= 0.1$ or < 0.5), then the motifs are removed less and the data is retained well.

• **Subgraph masking:** Masking the motifs with some specific features, masking the motifs with network communication motifs, protocol-specific motifs. It enhances the clustering process by specifically focusing on domain-specific variants. [equation (10)].

$$sm_i = \begin{cases} 0 & \text{with probability } \mathcal{P} \\ sm_i & \text{otherwise} \end{cases} \quad (10)$$

For each sample in the pretraining process, two motif hypergraphs Aug_{MH1} and Aug_{MH2} are formed. These hypergraphs leverage the co-occurrence and correlation between the data and statistical features of motifs, which are then embedded using a GCN to generate meaningful motif representations. The augmented motif hypergraphs Aug_{MH1} and Aug_{MH2} are given as input to the GCN layer, where the embeddings are constructed based on the Eq. (11).

$$Aug_{mhi}^1 = GCN(Aug_{MH1}), Aug_{mhi}^2 = GCN(Aug_{MH2}) \quad (11)$$

Where Aug_{mhi}^1 is the vectorization of motif (1) Aug_{mhi}^2 represents the vectorization of motif (2) The embedding is effectively used for capturing the relative relationship and feature correlations among the motifs considered. The Inter-Stage Contrastive Loss function works on the embedding space, using positive pairs (same motif with different augmentations) and negative pairs (unrelated motifs) to learn the robust representation. Equation (12) specifies the use of margin value to define the inter-stage loss and model to detect the transition between the stages.

$$Loss_{ISCL} = \sum_{a \neq b} \sum_{a \in Aug_{MH1}, b \in Aug_{MH2}} \max(0, \delta - \|Aug_{mhi}^1 - Aug_{mhi}^2\|_2) \quad (12)$$

Once the Inter-stage contrastive losses are specified, the motifs are selected based on the attention mechanism, and the highly relevant motifs are prioritized. The motif selection is based on the criteria such that the motifs with high centrality are selected and given higher importance. Then top-k motifs are selected and processed by the learning model. Equation 13 specifies the process of motif ranking. ς_m represents the higher topological relevance of motif graphs.

Motifs are ranked using an attention mechanism, prioritizing highly relevant ones based on importance. Frequently occurring motifs with high centrality are identified and given more importance. Top-k motifs with optimal paths are selected using criteria from Equation. (13), filtering out redundant or weakly contributing ones, focusing on relevant motifs for model training. ξ_m represents the frequency of occurrence of the motif with similar behavioral patterns. Θ_m represents the dimension of connectivity and relationship between the constructed motifs. With these factors, the top k motifs are prioritized.

$$\varrho_m = ((\varsigma_m) \cdot \log(\xi_m + 1)) / (\Theta_m + 1) \quad (13)$$

Each motif comprises nodes from the hypergraph, an embedding vector derived from the pretraining process, and co-occurring feature groups. These motifs are leveraged to enhance the GCN's ability to learn meaningful representations effectively. Then the layer-wise propagation process is initiated, where the motif hypergraphs are progressed using Equation (14).

$$MoGraM_{GCN}^{t+1} = \sigma((\Theta')^{-\frac{1}{2}} \left(A'((\Theta')^{-\frac{1}{2}} Aug_{mhi}^t \omega^t \right) \quad (14)$$

σ represents the activation function introducing the non-linearity to the model, A' adjacency matrix with added self-attention to capture both local and global dependencies, Θ' degree of the matrix A' , ω^t represents the trainable weights, and σ represents the activation function. GCN layers are initialized with embeddings learned via contrastive learning and motif features robust to missing data, noise and time drift (slow APT transitions). The multi-objective gradient moderator optimization function is used, where the loss functions considered are stage accuracy loss, stage transition loss, Motif importance loss, and inter-stage contrastive loss [Equation (15) to Equation. (17) and Equation. (12)]

$$S_{Loss} = - \sum_i Y_i \log(Y_i^{\wedge}) \quad (15)$$

Equation (15) specifies the stage accuracy loss of APT stages. Multi-stage accuracy is considered, and the loss value $\log(Y_i^{\wedge})$ is calculated.

$$ST_{loss} = \sum_{(i,j)} (1 - \cos(Aug_{mhi} - Aug_{mhj})) \quad (16)$$

Equation (16) imposes the temporal smoothness and the semantic continuity of the consecutive stages of APT. Aug_{mhi} and Aug_{mhj} represent the augmentation process of motif embedding for the features i and j . Cosine similarity measures the loss of similarity measures between the motifs in the embedding space. This loss also specifies that if the adjacent space embedding is with the value close to 1, then the motif stays close. If sudden embedding occurs, then it may be considered as noisy nodes and they are penalized.

$$M_{loss} = \sum_i \vartheta_i \cdot CCE(Y_i) \log(Y_i^{\wedge}) \quad (17)$$

Equation (17) assigns importance weights ϑ_i to the motifs that are most significant in identifying the stages of APT attack. $CCE(Y_i)$ Specifies the stage classification cross entropy loss where the motif i will appear. $Y_i^{\wedge}$ will performs the predicted probability of the APT stages. The total multi-objective loss function is calculated using the Eq. (18). Where the MoGraM employs adaptive reweighting and gradient projection to modulate task-specific gradients such as task loss sharpness, gradient conflicts and learning stability.

$$L_f = \lambda_1 S_{Loss} + \lambda_2 ST_{loss} + \lambda_3 M_{loss} + \lambda_4 Loss_{ISCL} \quad (18)$$

$$\nabla OGCN = \sum_{\kappa} \gamma_{\kappa} \cdot \nabla L_f \cdot \gamma_{\kappa} \quad (19)$$

Equation (18) specifies the overall loss function value summarizing all the losses, and Equation. (19) addresses sharpness, gradient conflicts, and stability concerns.

Proposed methodology

Sequential Analysis of Network patterns with Graph-based Learning (SANGL) approach integrates sequential analysis and graph-based learning analysis [Fig. 2]. Sequential analysis represents the Pyramidal Principal Component Analysis (PyPCA) for extracting progressive network traffic APT patterns. Graph-based learning represents the process of constructing motif hypergraphs for analysing subtle attack patterns in APT, and Graph Convolution Network (GCN) is used to correlate APT attack stage-wise transitions. Initially, the optimal features are extracted, then it is transformed into motif-based hypergraphs where nodes represent entities or events, and

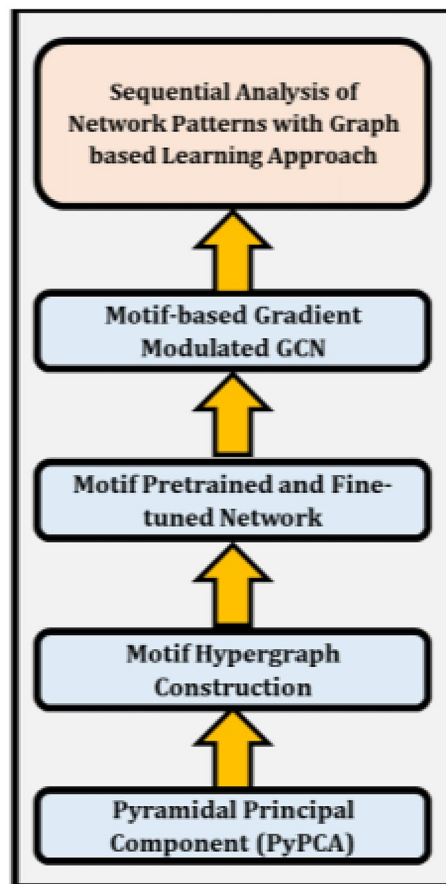


Fig. 2. Sequential Operation of the proposed SANGL approach.

hyperedges encode temporal dependencies and contextual relationships. The hypergraph is input to a Graph Convolutional Network (GCN), which learns intra-stage and inter-stage dynamics. An attention mechanism assigns higher weights to motifs with strong sequential continuity and structural significance, enabling SANGL to model sequential evolution and relational topology of APT behaviour.

Data collection

To validate the efficiency of the proposed approach, the CICAPT IIoT Dataset was used⁴⁶, which was simulated by the Canadian Institute of Cybersecurity in an IIoT-enabled environment. The simulation testbed was designed using the Brown-IIoTbed architecture, utilising the NS3 simulator tool on an Ubuntu host and comprising two Ubuntu and two Kali Linux virtual machines, two Raspberry Pi devices, and two IoT sensors to replicate the real-time IIoT environment. A cumulative of 20 various APT groups are analysed, and the specific attack techniques are gathered using the MITRE ATT@CK framework for analysis. The respective TTPs, along with the stages are captured in the Fig. 3. The dataset's features enable detection of specific techniques like Command & Control and Persistence, facilitating end-to-end traceability from feature-level activity to TTPs and effective APT detection. Data collection occurs in two phases: Benign and attack Phases. The system log flows and network flows captured during the process and mentioned elaborately in the Section 5.

Figure 4 represents the detailed working of the proposed approach, which consists of three phases: (i) Pyramidal Principal Component Analysis (PyPCA) for extracting the essential data for modelling, (ii) High fidelity motif graphs model to capture the higher order relationship of APT attack patterns and (iii) Motif Hypergraph-based Pretrained and Gradient-Optimized Graph Convolution Network for APT Detection.

Data pre-processing

In the IIoT environment, the large volume and complexity of data acquisition lead to null values, which impair the performance of learning models due to increased noise and reduced quality. Therefore, pre-processing is crucial in mitigating these data degradation issues. The proposed SANGL approach includes four data preprocessing methods: Feature Scaling, Data Transformation, Sampling and Normalization. The observed dataset includes large values which will degrades the performance of the learning model leads to biased prediction hence Feature scaling is applied to ensure all data samples are on a consistent scale. Following, data transformation process is applied to identify the uniqueness between features and then combined based on their significant values. Since APT involves multiple stages, it generates a large amount of data with redundant information, making it challenging to identify underlying attack patterns. To address this, the Data Dominance mitigation method is used to reduce the number of benign samples that outnumber attack samples. Finally, data normalization process is performed using minmax normalization process.

Pyramidal Principal Component Analysis (PyPCA)

After performing these pre-processing techniques, to extract the optimal features and to reduce the data dimension, the PyPCA method is used. The PyPCA method is employed to extract the variance-aware features,

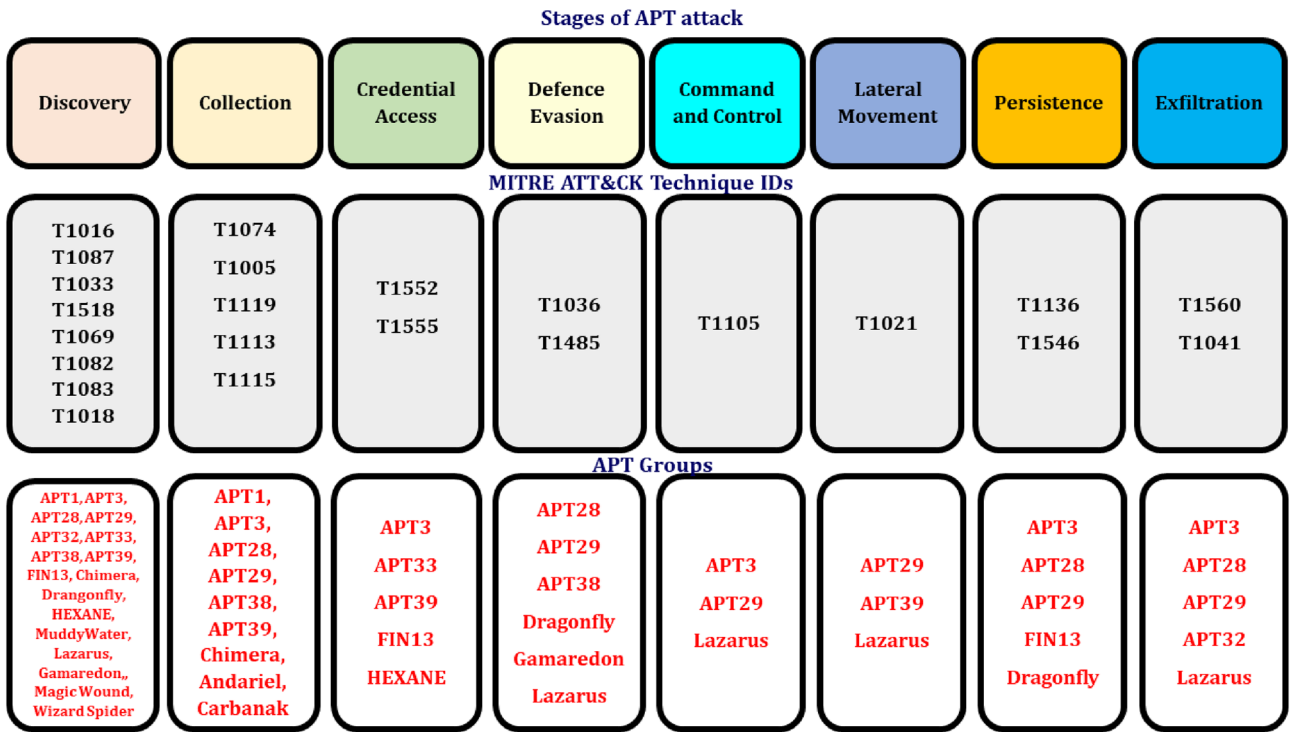


Fig. 3. Dataset overview.

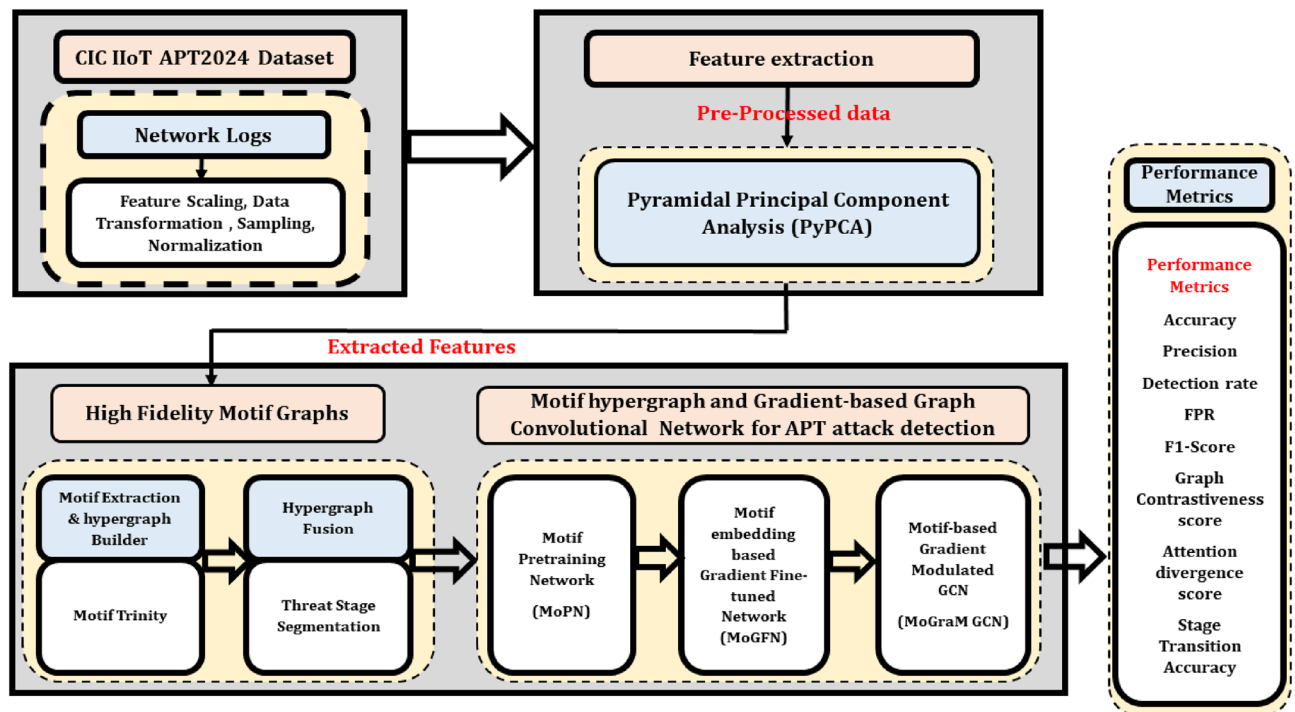


Fig. 4. Working architecture of SANGL approach.

to capture subtle, crucial attack variants of APT attack stages, leading to more precise predictions of stage transitions and enhanced threat detection capabilities. It has a pyramidal structure using three methods: (i) Entropy-based weighting method – to remove the uncertainty in the observed data, (ii) Correlation-based weighting method – to select the interdependencies with the features and (iii) Temporal-based weighting method – to select the time-sensitive patterns for stage-wise APT detection. Each method is assigned weights, for which the temporal method is given higher weights to capture the time-based features that contribute more towards the APT stage transition analysis Algorithm 1 Fig. 5. Steps involved in PyPCA,

Step 1 Compute weights for each feature using entropy, correlation, and temporal methods. The entropy method assigns higher weights to features with lower entropy. The correlation method assigns lower weights to redundant features. The temporal method assigns higher weights to features with high mutual information with time. [Equation (1) to Equation (3)].

Step 2 The hyperparameters of the three methods are optimized using the Bayesian optimization model. The weights are tuned dynamically based on the importance of each feature, and the final weights are combined. The model ensures that the combined weights are equal to the value of one. Equation (4) & Equation (5).

Step 3 The combined weights are then scaled to ensure constant variance for the entire feature to remove the dominance.

Step 4 Once the features are scaled, the covariance matrix is formed to capture the relationship between the obtained features.

Step 5 The PyPCA method includes the attention mechanism to rank the features computed via the covariance matrix, using the attention mechanism and ensuring the most important features are selected. [Equation (6)]

Step 6 Finally, the optimal principal component features obtained with reduced dimensions are considered for analysis.

Motif hypergraph-based pretrained and gradient-optimized GCN for APT Detection

This phase emphasizes the construction of motif hypergraphs followed by pretraining process, then fine-tuning using an attention mechanism and an optimized Graph Convolution Model for correlating the stages of APT. These processes are captured in two sections: (i) High Fidelity Motif Graph construction, and (ii) Motif hypergraph and Gradient based graph convolution network.

High-fidelity motif graphs

Motif Trinity and Threat Stage Segmentation.

Input: IIoT network data D, f_i
Output: Extracted data after PyPCA

Data Pre-processing
 Remove features with zero variance
 For each feature f_i :
 If variance == 0 or all the values of $f_i == 0$
 Remove f_i

Data Transformation
 For each feature f_i in D :
 Apply feature scaling transformation = $\frac{f_i - \min(f_i)}{\max(f_i) - \min(f_i)}$

Data dominance mitigation $B' = B \cdot (\sum_{i=1}^n |A^i|) / (B \cdot \beta)$

Pyramidal Principal component Analysis (PyPCA)
 do
 $weight_{Ei} = 1 - f_i / \max(f_i)$
 $weight_{Ci} = 1 - \frac{1}{n} \sum_{n=1}^1 |corr(f_i, f_n)|$
 $weight_{Ti} = MI(f_i: ts)$
 Combine all the feature weights $hybridweights = \omega_E weight_{Ei} + \omega_C weight_{Ci} + \omega_T weight_{Ti}$
 Weighted data matrix construction $Weighted\ Samples = Samples * hybridweights$
 Apply $Sd = Weighted\ Samples - \mu / \sigma$
 Compute $C_M = Sd^T \cdot diag(Weighted\ Samples) \cdot Sd$
 Compute $Q = C_M * W_q, K = C_M * W_k, V = C_M * W_v$
 Apply Attention (s_d) = $P_k Weighted\ Sample ((P_k)^T s_d)$
 Reduced data dimensions and optimal features extraction
 End

Motif Trinity
Input: Attention s_d
Output: T_{GM} (Triangular Motif)
 Start
 Initialize $G = \mathbb{V}, \mathbb{E}; u, v, w \in \mathbb{V}$, empty list T_ℓ
 If $(u, v), (u, w), (v, w)$ in $\mathbb{E}$
 Then concatenate u, v, w in ℓ
 Return T_{GM}
 End

Input: Attention s_d
Output: C_{GM} (Clique Motif)
 Start
 Initialize $G = \mathbb{V}, \mathbb{E}$; empty list c_ℓ
 If each node v in $\mathbb{V}$
 Do $N(v) = neighbours\ of\ v$
 For each subset S of $N(v)$ with size ≥ 3
 Check all the nodes are connected
 If yes then add v
 Return C_{GM}
 End

Input: Attention s_d
Output: B_{GM} (Bipartite Motif)
 Start
 Initialize $G = \mathbb{V}, \mathbb{E}$; empty list B_ℓ ; set a ; set B
 If interaction exists between groups then add edges to B_{GM}
 Return B_{GM}
 End

$HG_{APT} = \mathbb{V}, \mathbb{E}$

Threat Stage Segmentation
 Compute Equation (8) to Equation (10) for motif augmentation process
 Compute Aug_{mhi}^1 and apply $Loss_{ISCL}$ for Motif augmentation and Inter-stage contrastive loss
 Perform q_m for ranking the motifs
 Compute Equation (13) and Equation (14)
 Compute fine-tuning of GCN using Equation (15) to Equation (17) considering the loss function
 end the process with Transition of APT stages from the Equation (18) and Equation (19)

Algorithm 1. Working flow of SANGL approach.

The optimal features are extracted using the PyPCA method, and the hypergraphs are formed with $HG = \mathbb{V}, \mathbb{E}$ where $\mathbb{V}$ represents the vertices (Features) and $\mathbb{E}$ represents the edges (relationship between the features based on the stages). To discriminate the discrete behavior of APT attack patterns, the motifs are formed based on motif trinity concepts: (i) Triangular Motif, (ii) Bipartite Motif and (iii) Clique Motif.

1. **Triangular motifs (T_{GM})** capture the interrelated patterns or cyclic paths between the features, where the set of three nodes is interconnected in a cyclic manner, if the same protocol is used for communication by distinct IPs, then it is flagged as a triangular-based motif.

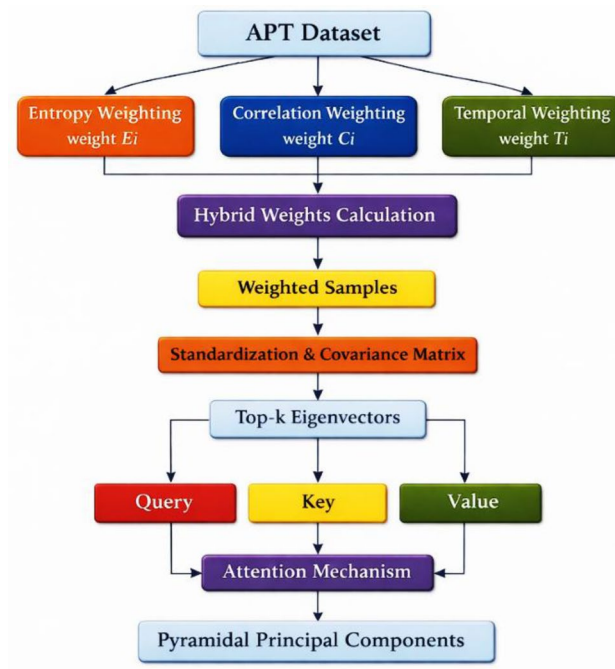


Fig. 5. Flowchart for the working of PyPCA method.

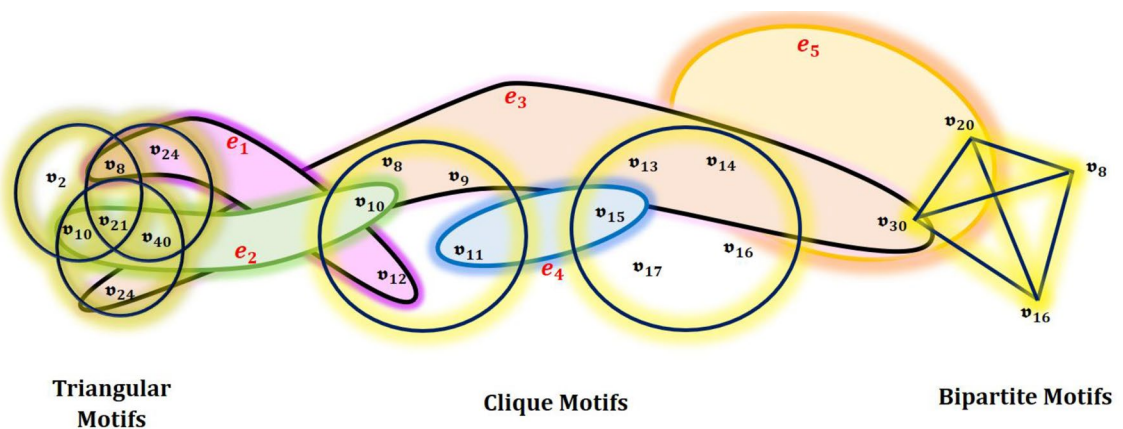


Fig. 6. Motif Trinity Process.

2. **Bipartite motifs** (B_{GM}) are formed based on the two disjoint nodes, where the connections are made between the set of nodes and not within the nodes. If the internal IP is the source IP, and it is connected to the external destination IP through specific ports, then the communication is flagged as a bipartite motif.

3. **Clique motifs** (C_{GM}) specifies the interconnection between each node. i.e., Suppose five features are considered as nodes, such as Source IP, Destination IP, protocol type, Syn_count, and Ack_Count. In that case, there exists an interconnection between these nodes, and it is categorized into clique-based motifs.

Based on these criteria, the motif hypergraph is constructed as $HG_{APT} = \mathbb{V}, \mathbb{E}$. $\mathbb{V}$ represents the nodes and $\mathbb{E}$ represents the hyperedges. Motif trinity is considered a hypernode $\mathbb{V}_0 to \mathbb{V}_n$ where $\mathbb{V}_n$ represents the number of optimal features extracted in the previous phase using PyPCA. The stages of APT are considered as hyperedges and represented as $e \in \mathbb{E}$ where e is regarded as the stages of APT. Once the motifs are framed, it is described as $HG_{APTd} = HG_{APT}$. The detailed workflow is depicted in algorithm 1 and Fig. 6.

From the above figure, it is inferred that the motifs are formed using the extracted optimal features. e_1, e_2, e_3, e_4 and e_5 represents the edges which is the stages of APT and the nodes are grouped based on the motif trinity process and to differentiate among the stages the color variations are included. The edges are formed based on the stages of the APT. The node e_3 is the lateral movement stage which encompasses the features ($v_{24}, v_{10}, v_8, v_9, v_{13}, v_{14}, v_{30}$) observed from all the motif types.

Motif hypergraph and gradient-based graph convolutional network

Following the construction of motif hypergraphs, the pretraining process is initiated, where motif augmentation and loss function play pivotal roles in generating robust motif-level embeddings for precise identification of APT stages Fig. 7.

Motif pretraining network (MoPN)

The pretraining network composes of augmentation process where it is achieved using three techniques namely motif temporal drift, motif dropout and subgraph masking method. This structure defines a complete diverse graph representation of APT attack patterns. The loss function is constructed based on the inter-stage contrastive loss. This component maximizes the similarity between embeddings of the same motif under different augmentations, while minimizing the similarity between embeddings of other motifs. Equation (12).

- **Motif Temporal Drift** method introduces time delays between sequential events in each motif instance, simulating real-world variability in event timing Equation. (8). By including temporal drift, the model becomes more robust and learns to identify the motifs even if their timing differs. In Fig. 7, the motif pretraining phase shows a sample temporal motif, colored in yellow, which was eliminated due to variations in attack patterns. At the emergence of time delays, the motif behavior changes, showing how the APT attack adapts.

- **Motif Dropout** is used to randomly drop motifs based on the probability it calculates on Equation. (9), effectively performing sparse log simulation or injecting noise into the system. By using motif dropout, the model learns from complete data by enhancing its resilience against missing information and improving its overall performance. From Fig. 7, it can be noticed that the motif having a probability value of 0.5 is removed as it has sparse logs, degrading the learning model's performance.

- **Subgraph Masking** method involves masking specific features or motifs, such as network communication or protocol-specific motifs Equation (10). By selectively masking certain subgraph components, the model is encouraged to focus on the most relevant and informative features, reducing its reliance on any single feature or motif and promoting more robust representation learning. From Fig. 7, it is inferred that the subgraph masking is presented in dark shades, as specific nodes related to particular protocols, such as network-based protocols and communication-based protocols, are purposefully removed, and the remaining motifs are considered for processing.

Once the motifs' hypergraphs are refined using the augmentation process Equation (11), each sample generates two augmented motif hypergraphs, Aug_{MH1} and Aug_{MH2} , which capture co-occurrence and correlation patterns between motifs. These hypergraphs are fed into a Graph Convolutional Network to produce significant motif representations that effectively encode contextual relationships and feature correlations, providing a deeper understanding of the underlying patterns and motifs. Sequentially, the Inter-Stage Contrastive Loss function is employed to optimize motif representations. It finds the similarity between augmented versions of the same motif while maximizing differences between distinct motifs. This process enhances the model's ability to identify stage transitions by learning robust and discriminative representations Equation (12). In Fig. 7, the yellow color highlighted temporal drift process represents the nodes which is removed, likewise in motif dropout

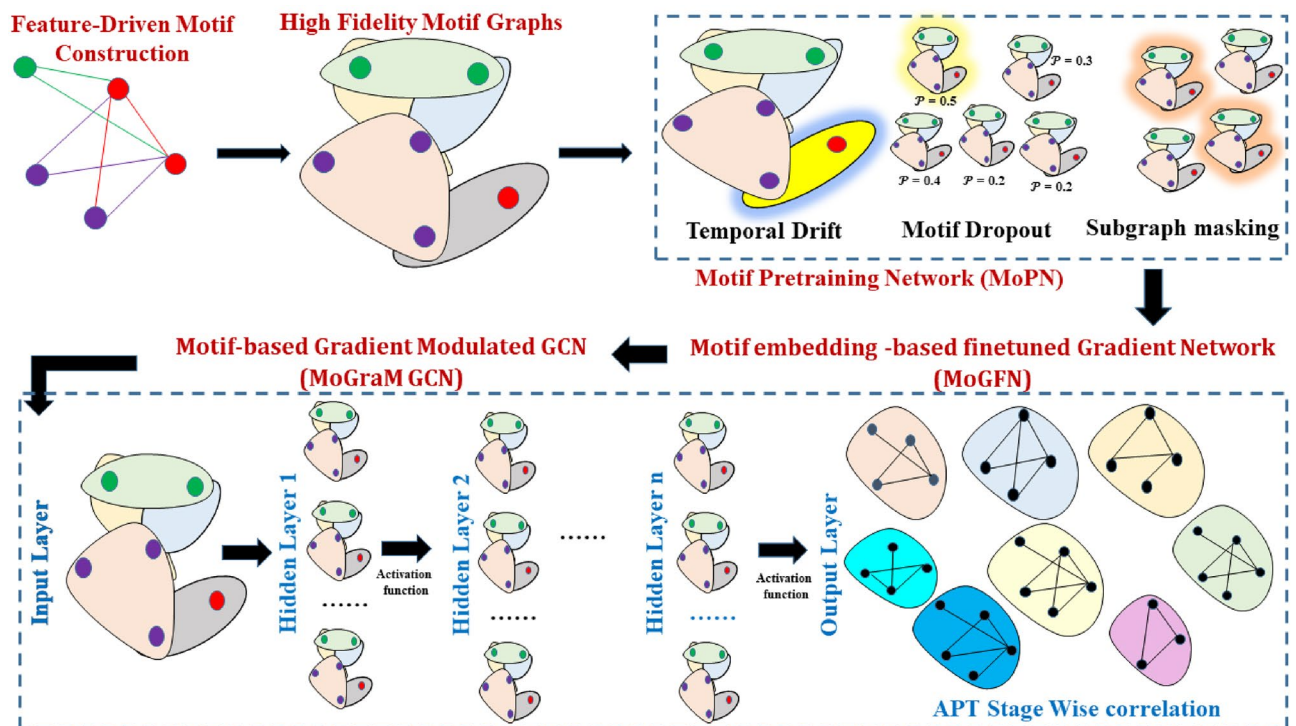


Fig. 7. Architecture of Motif trinity - MoGraM based Optimized GCN.

the highlighted motif with p value 5 is removed. Subsequently, in subgraph masking the hypergraphs 1 and 4 is removed.

Motif embedding-based gradient fine-tuned network (MoGFN)

MoGFN module further improves the motif embeddings through a gradient fine-tuning process, ultimately yielding more robust and accurate representations for APT detection. MoGFN module further improves the motif embedding through a gradient fine-tuning process, ultimately yielding more robust and accurate representations for APT detection. The attention mechanism is proposed which presents then a multi-dimensional weighting module with three variants such as structural significance, frequency of occurrence and temporal relevance. In structural significance, the motifs with higher connectivity and influencing the hypergraph are assigned with higher weights for top-k motif selection. The motifs that frequently appears across the sequential stages of APT attack are prioritized to capture the behavioral dependencies. The motifs occurs with the strong temporal continuity are captured to preserve the APT attack progression. Each are assigned with motif values which are then normalized to rank the motifs and top-k motifs are selected as discriminative structural units. Then Top-k motifs ϱ_m is used as input features for an optimized GCN, and then a layer-wise propagation process is initiated, where motif hypergraphs are progressed using trainable weights and activation functions, enabling the model to learn meaningful representations and detect APT attacks effectively. This allows the GCN to distinguish malicious motif views from benign ones and generalize better to unseen but semantically similar attacks, which is crucial for detecting APTs in IIoT environments attack behaviors are often incomplete or obfuscated Equation (14).

Motif-based gradient modulated GCN (MoGraM GCN)

GCN typically optimizes a single loss function, such as classification loss, which often leads to suboptimal convergence and gradient interference. However, APT detection is a multi-stage specific problem requiring optimization of multiple objectives, including stage classification, anomaly detection, and attack prediction, necessitating a more nuanced approach to loss function design. Considering this, a Multi-Objective Loss Function is designed which optimize stage accuracy loss S_{Loss} [Equation (15)], stage transition loss ST_{Loss} [Equation(16)], motif importance loss M_{Loss} [Equation(17)], and inter-stage contrastive loss $Loss_{ISCL}$ Equation (12) that will adaptively balance the gradients from different tasks to achieve consistent optimization. This allows the model to balance and optimize multiple tasks, ensuring that balanced learning across all the tasks numerically increases the detection accuracy and adaptability towards the unseen behaviour of APT attacks. Then, the process of gradient moderation is brought into place, where task-specific gradients are modulated using adaptive reweighting and gradient projection with respect to sharpness, gradient conflicts, and stability concerns [Equation (18) & Equation. (19)]. This ensures a stable and efficient learning that can effectively detect the APT stage-wise transitions in IIoT environments.

Experimental results

This section outlines the dataset used for validation of the proposed SANGL approach, experimental configurations, performance metrics, and observations.

Dataset description

The CICAPTIIoT dataset⁴⁶ simulated in an IIoT environment was used for the validation of the proposed SANGL approach. The dataset comprises various APT groups, from which various attack patterns are derived to simulate the stages of an APT attack. The bias of using this dataset for observation specifies that this has various attack stages. The observed CICAPT IIoT 2024 dataset⁴⁶ was simulated in an IIoT environment consisting of several APT attack techniques from various APT groups in two phases: Phase 1 has 46,773 nodes and 12,103,705 packets with benign samples, Phase 2 has 52,954 nodes and 9,535,819 packets with all the attack variants of APT attack. A total of 1004 attack samples are observed with 70 features extracted using the CIC flow meter tool. 85 features are derived. The detailed analysis of attack states is mentioned in the table below Table 3.

Phase 1		Phase 2	
Label	Count	Label	Count
Benign	12,103,705 no. of. packets	Benign	9,535,819 no. of. Packets
		Collection	460
		Exfiltration	42
		Credential Access	58
		Defence Evasion	192
		Discovery	138
		Persistence	44
		C&C	56
		Lateral Movement	14

Table 3. Dataset description with numerical values.

In addition to the primary dataset, three more datasets: DAPT20⁴⁸, Unraveled⁴⁹, and Edge-IIoT⁴⁷ were utilized for validation to strengthen the performance of the proposed approach further and ensure its robustness and generalizability.

Dataset significance

The CICAPTIIoT dataset provides a diverse set of flow-level and protocol-specific features of precise value in the modelling of the transition among APT stages, such as flow_duration, ack_flag_number, syn_flag_number, Rate, Srate, and Tot size. These features capture both temporal dependencies, including duration and inter-arrival times, and behavioural dynamics, such as flag-based signalling, rate fluctuations, and control flow patterns, all are essential for distinguishing the different phases of APT stages. Beyond the detailed multistage labelling in CICAPTIIoT dataset, it offers a well-designed and benchmark-ready structure for the scalability of adaptive detection frameworks within heterogeneous IIoT networks. This dataset captures traffic-pattern variations from different industrial protocols, device communication behaviours, and time variations from multiple APT campaigns, all of which make it ideal for model analysis in terms of robustness, noise resilience, scalability and cross domain transferability under dynamic network conditions. The large volume of flow records, along with the balanced distributions of benign and APT attack instances, enables reliable training of the proposed approach to avoid overfitting to particular stages or network nodes. Integrated with other datasets such as BoT-IoT⁶¹, ToN-IoT⁶², DAPT20⁴⁸, Unraveled⁴⁹ and EDGE-IIoT⁴⁷, the observed CICAPTIIoT dataset serves as a baseline dataset for cross-domain generalization, allowing to evaluate the scalability, and resilience of APT detection models in real-world IIoT environments.

Working environment

The entire experimentation was carried out in the system with 16GB primary memory was used with AMD Ryzen 7 8845HS processor 780 M Graphics card is used. Python 3.13 version is used for programming perspectives. Libraries such as NumPy, Pandas, Keras and TensorFlow with High Performance Computing (HPC) support and GPU acceleration is used to enhance the processing power and to reduce the latency of the proposed approach training process.

Performance Metrics

The proposed approach's proficiency is validated using various metrics, including accuracy, precision, F1-score, detection rate, and False Positive Rate (FPR) Equation (20) to Equation. (24). Graph Contrastiveness Score, Attention Divergence Score, and Stage Transition Accuracy are used to assess the performance of the proposed Motif graph approach. The overall evaluation is proceeded with the metrics such as True Positive (τ_p), True Negative (τ_n), False Positive ($\mathcal{F}_p$) and False Negative ($\mathcal{F}_n$).

$$Accuracy = \frac{\tau_p + \tau_n}{\tau_p + \tau_n + \mathcal{F}_p + \mathcal{F}_n} \quad (20)$$

$$Precision = \frac{\tau_p}{\tau_p + \mathcal{F}_p} \quad (21)$$

$$Recall \text{ or } DetectionRate = \frac{\tau_p}{\tau_p + \mathcal{F}_n} \quad (22)$$

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (23)$$

$$FalsePositiveRate = \frac{\mathcal{F}_p}{\mathcal{F}_p + \tau_n} \quad (24)$$

Graph Contrastiveness score defines the embeddings that are separated with the distinct APT behaviours Equation (25). Contrastiveness refers to the analysis of differences in the degree of nodes. The score is high, indicating that the inter-class compactness is higher between the classes.

$$\text{Graph Contrastiveness Score} = 1/|\wp| \sum_{(i,j) \in \wp} sc(h_i, h_j) - 1/|\aleph| \sum_{(i,\aleph) \in \aleph} sc(h_i, h_{\aleph}) \quad (25)$$

Where $\wp$ defines the positive classes, $\aleph$ represents the negative classes, sc defines the cosine similarity function, h_i, h_j embedding of the same class and $h_{\aleph}$ embedding of a different class. Attention divergence score is used to define how the model assigns weights to different motifs [Equation (26)].

$$AttentionDivergenceScore(\mathfrak{s}, \mathfrak{s}') = Divergence_{KL}(A_{\mathfrak{s}} || A_{\mathfrak{s}'}) = \sum_{i=1}^{\eta} A_{\mathfrak{s}}(i) \log \left(\frac{A_{\mathfrak{s}}(i)}{A_{\mathfrak{s}'}(i)} \right) \quad (26)$$

where $\mathfrak{s}, \mathfrak{s}'$ defines the stages, $A_{\mathfrak{s}}$ and $A_{\mathfrak{s}'}$ defines the attention distribution for the stages. $A_{\mathfrak{s}}(i)$ defines the attention score distributed for the motif i in the stage $\mathfrak{s}$. Stage Transition Accuracy measures the accuracy of the transition between stages in a sequential detection process. Equation (27)

$$Stage\ transition\ accuracy = \frac{No.\ of\ correct\ stage\ transition\ detected}{Total\ No.\ of\ actual\ Transition} * 100$$

(27)

Results and Discussion

The proposed SANGL approach is validated in various aspects and the appropriate observations obtained for each research observations are captured in this section. Table 4 presents the elaborated view of parameter setting for the proposed SANGL approach followed by the discussions. The network employs GELU (Gaussian Error Linear Unit) activations to handle dynamic data variants. The network makes use of GELU activations to work with dynamic data variants. A dropout rate of 0.4, weight decay of 1e-5, and gradient clipping of 1.0 are used to enhance non-linearity while mitigating overfitting, in conjunction with a learning rate of 0.001, governed by a OneCycleLR learning rate scheduler to ensure stable and adaptive convergence over 200 epochs with a batch size of 512 and with the 10-fold cross-validation as random feeds for the SANGL approach.

Initially, the process begins with dataset observation, as the dataset has large values of traffic volumes when comparing two phases of data, hence feature scaling is performed. Normalization is performed to remove the outliers. To construct motif hypergraphs, encoding plays a vital role as it encodes the features with same id host with the same embedding. After pre-processing, the data dominance mitigation method is performed, where the sample distribution after preprocessing is captured in Fig. 8. Finally, for further processing, 225,000 samples are obtained containing both benign and attack samples in equal distribution.

Research Observation 1 How effectively PyPCA method enhance the identification of critical APT behavioural patterns when compared to conventional feature extraction techniques?

The PyPCA method is effectively used to extract the optimal features with a count of 48, enabling focused learning of relevant patterns. The process begins with feature pruning to eliminate null values, followed by data dominance to mitigate class imbalance by down-sampling the benign instances, which initially dominated the attack samples. PyPCA was applied with weighted importance scores of 0.3, 0.4, and 0.4, respectively, to extract significant features. Figure 9 shows that the features extracted with PyPCA give insight into important characteristics of network traffic patterns relevant to APT attacks in the IIoT environment, such as network protocol features like TCP, UDP, HTTP, and SSH; traffic pattern features, sum_duration and average_duration; packet-level features, ack_count and syn_flag_number, among others; and statistical features, rate and variance. Protocols such as TCP, UDP, LLC, IAT, DNS, ARP, MAC, HTTP, and ICMP are the basis necessary in IIoT for efficient, reliable, and secure communication among devices. Each of these features contributes to the detection of potential stages of APT attack, including lateral movement, command and control communication, data exfiltration, and network reconnaissance, in order for the model to identify unusual patterns and anomalies indicative of APT attacks in IIoT environments.

Research Observation 2 Does motif hypergraph clustering improve the modelling of higher-order inter-stage relationships in multi-stage APT attacks?

Parameter	Values
Input Sample Size	225,000 Samples
Class Labels	Nine (Benign, Collection, Exfiltration, Credential Access, Defence Evasion, Discovery, Persistence, Command and Control (C&C), and Lateral Movement)
Feature dimension after PyPCA	48
Activation Function	GELU
Dropout Rate	0.4
Learning rate	0.001
Learning rate scheduler	OneCycleLR
Batch Size	512
Epochs	200
GCN Aggregator	Mean Aggregator with MoGraM modulator
GCN Layers	4
Gradient Modulation	Adaptive weighting via entropy-based backpropagation
Optimizer	MoGraM
Weight factor for motif matrix	0.5
Weight of stage transition	0.4
Motif embedding size	64
Weight initialization	Kaiming
Gradient Clipping	1.0
Weight decay	1e-5

Table 4. Parameters and their values of the proposed approach.

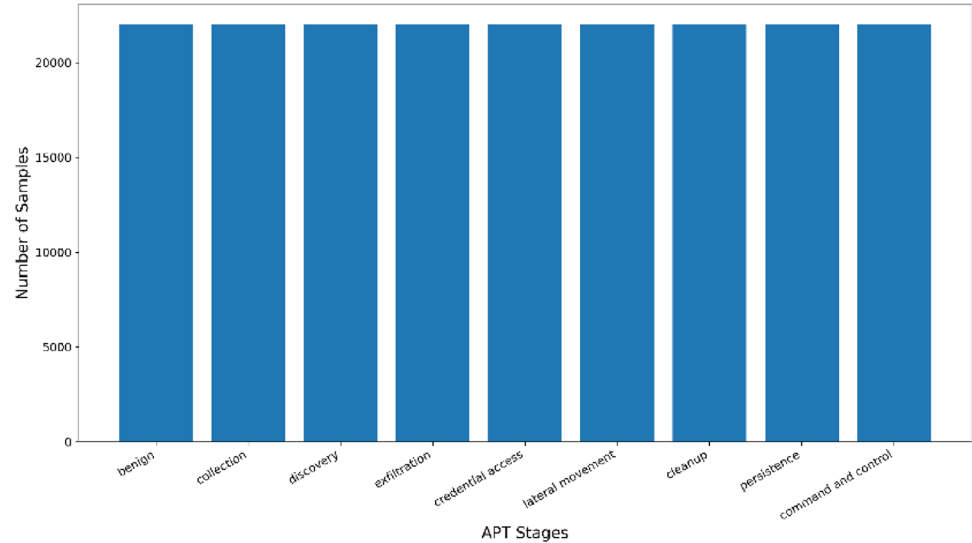


Fig. 8. Visualization of the Sample distribution using data dominance mitigation method.

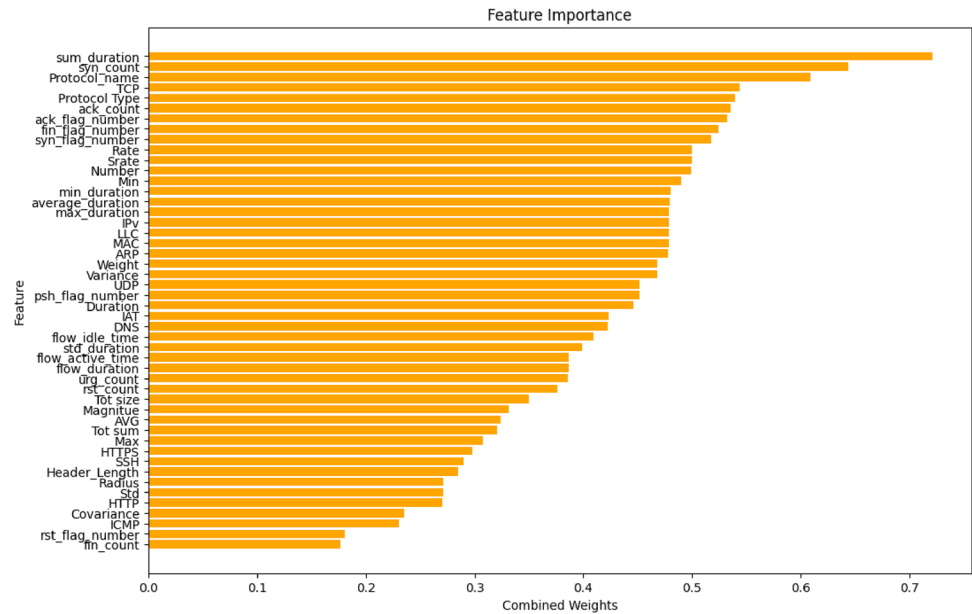


Fig. 9. Optimal features extracted from the PyPCA Method.

'e ₀ ':	{'V31', 'V15', 'V29', 'V28', 'V30', 'V26', 'V37', 'V34', 'V42', 'V43', 'V44', 'V39', 'V40'}
'e ₁ ':	{'V48', 'V6', 'V2', 'V44', 'V12'}
'e ₂ ':	{'V14', 'V15', 'V16', 'V19', 'V9', 'V12'}
'e ₃ ':	{'V34', 'V37', 'V36', 'V7', 'V8', 'V9', 'V4', 'V23', 'V3'}
'e ₄ ':	{'V26', 'V28', 'V30', 'V41', 'V12'}
'e ₅ ':	{'V22', 'V45', 'V21', 'V4', 'V44'}
'e ₆ ':	{'V21', 'V48', 'V14', 'V5', 'V3', 'V41'}
'e ₇ ':	{'V34', 'V36', 'V9', 'V23', 'V44', 'V12', 'V19'}
'e ₈ ':	{'V36', 'V38', 'V23', 'V5', 'V10', 'V11', 'V35'}

Table 5. Hyperedges and Hypernodes of motif hypergraph.

Yes, the motif hypergraph improves the modelling process by segmenting the motif construction into three types, which enables the model to capture intricate attack patterns and their relation effortlessly. The efficiency of the Motif hypergraph is evaluated using (i) t-SNE visualization of clusters, (ii) Inter and Intra stage separability, (iii) significance of motif augmentation process. Primarily, the process begins with the construction of hypergraphs for which the hyperedges and hypernodes are obtained and depicted in Table 5.

These features are considered as the nodes, and the labels are regarded as edges, the features are labelled from $\mathbb{V}_1$ to $\mathbb{V}_{48}$ and edges are considered as e_0 to e_8 . All these features are mapped to the motif trinity process, and the motifs are formed. Then the motif hypergraph is constructed using the motif trinity process. E.g., the node e_0 refers to the benign samples with the nodes $\mathbb{V}_{31}, \mathbb{V}_{15}, \mathbb{V}_{29}, \mathbb{V}_{28}, \mathbb{V}_{30}, \mathbb{V}_{26}, \mathbb{V}_{37}, \mathbb{V}_{34}, \mathbb{V}_{42}, \mathbb{V}_{43}, \mathbb{V}_{44}, \mathbb{V}_{39}$ and $\mathbb{V}_{40}$ where it represents flow_duration, average_duration, std_duration, flow_idle_time, flow_active_time, IAT, Tot sum, Tot size, Radius, Std, HTTP and HTTPS. These entire features relate to the legitimate user behavior, i.e. HTTPS, which relates to the trusted domain connection, and IAT belongs to the normal behavior.

Figure 10 visualized t-SNE results of the CICAPT-IIOT dataset. It illustrates the motif hypergraph clusters before and after the pretraining process. Before pretraining, the data points exhibit significant overlap, resulting in poorly defined clusters for motif embeddings. This overlap leads to inter-stage confusion, failing to capture the underlying patterns of APT attacks. Consequently, the model's ability to learn stage transitions and distinguish motifs across different APT stages is compromised, yielding suboptimal performance. After pretraining, the right side of the figure shows that motif embeddings become highly discriminative across all labels, forming distinct high-density clusters. This superior clustering enables the downstream task of optimized performance in GCN, allowing for better learning and classification. The yellow lines correspond to the tightly coupled motif clusters, which show a high degree of overlap before pretraining (left figure). These points to poor separation of APT attack stages in IIoT ecosystems. In contrast, after pretraining, clusters become more distinct with reduced overlap, indicating that stages of APT result in better separation. Black dots demonstrate how clusters switch from loose to tight coupling, evidencing an improved capability of the model for complex APT pattern identification and distinguishing in IIoT environments. This allows for enhanced detection and thus analysis of the APT attacks. The values of the silhouette score before and after pretraining are given as $SS_i = 0.19$ and 0.71 , respectively. The first value, before pretraining, demonstrates the fact that the cluster is high with weakly separated motifs for APT stage correlation. The second value refers to the fact that there is high cohesion within APT stages and high inter-stage separation.

Figure 11 shows how margin selection affects APT detection performance, looking at metrics like Normalized Accuracy, Inter-Stage Separability, and Intra-Stage Similarity. A margin of 0.1–0.5 balances these metrics well, helping the model distinguish between attack stages while keeping similarity within stages. Inter-Stage Separability stays stable, dipping slightly at higher margins, suggesting a trade-off between separation and overfitting. Intra-Stage Similarity peaks at a margin of 0.2–0.5, dropping off at larger margins. A margin of 0.1–0.5 works best, letting the model learn robust APT tactic representations and boosting detection. Outside this range, performance drops due to under-training or over-penalization, hurting detection of complex APT attacks.

Motif hypergraphs in Fig. 12 illustrate the feature importance and anomalies in APT events, a feature identifier followed by its magnitude represents each edge. The values, such as $\mathbb{V}_{29} = 4.4935$ and $\mathbb{V}_{15} = 1.1948$, indicate strong deviations, potentially corresponding to unusual network activities, while values $\mathbb{V}_3 = 0.0818$ and $\mathbb{V}_9 = 0.1437$ shows malicious behaviour. The diversity in node values highlights the heterogeneous nature of APT events, and these labelled values enable the capture of fine-grained anomalies, updating motif-based learning models. For model generalization, the motif augmentation process is carried in the pretraining phase using three techniques.

The above and the below figures illustrates motif-based learning techniques in IIoT networks, including motif temporal drift, motif dropout, and subgraph masking. Motif temporal drift [Fig. 13] simulates temporal inconsistencies through nodes like $\mathbb{V}_{12}$ and $\mathbb{V}_{37}$ reappearing with slight variations, mimicking attackers' tactics to evade detection. Subgraph masking suppresses central nodes ($\mathbb{V}_3, \mathbb{V}_5, \mathbb{V}_{48}$) and amplifies peripheral ones

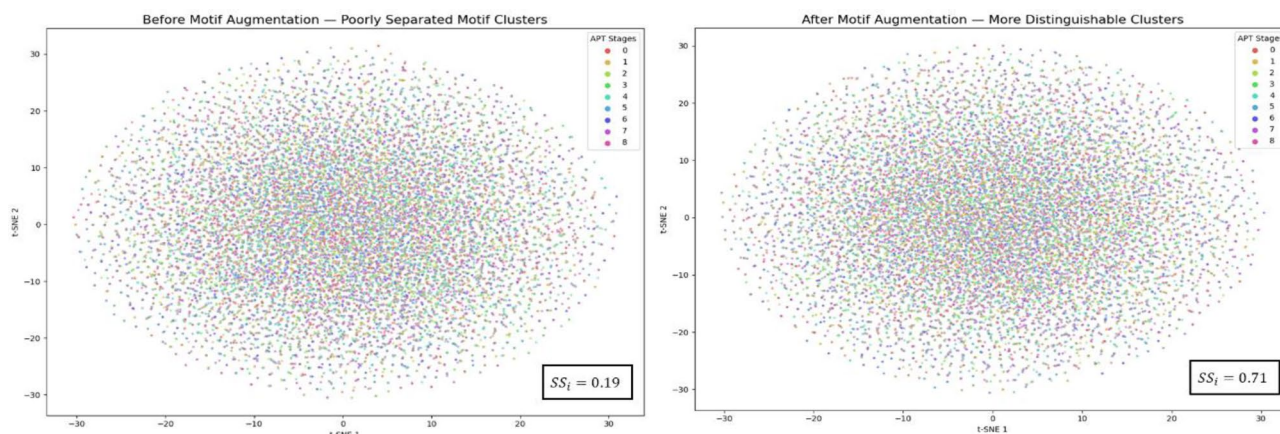


Fig. 10. t-SNE visualization of motif graphs before and after augmentation process.

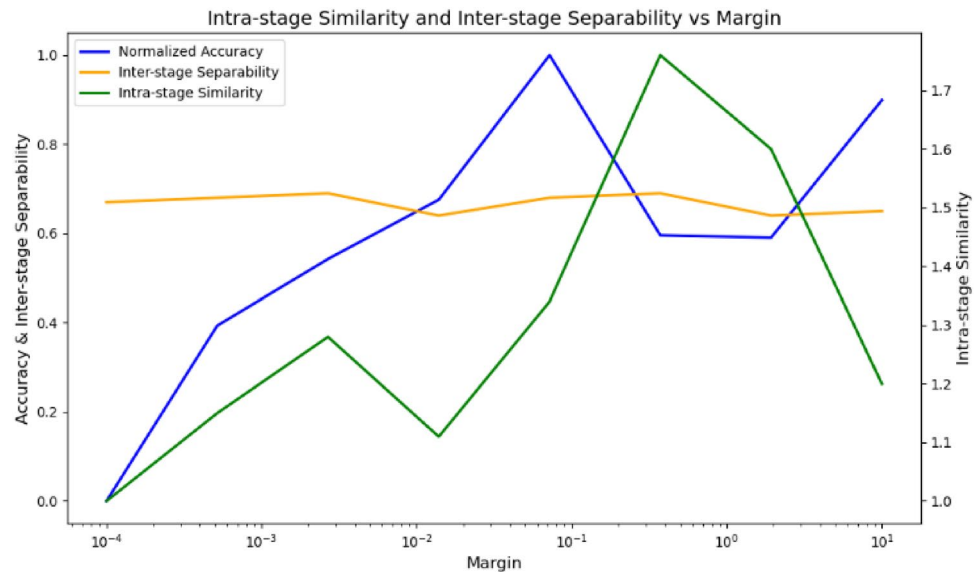


Fig. 11. Effect of Margin Selection on Inter- and Intra-Stage Similarity.

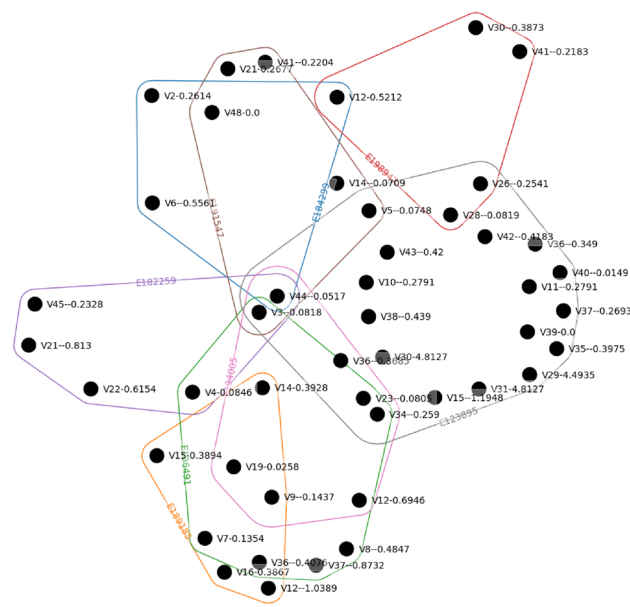


Fig. 12. Motif hypergraph covering all nodes and edges.

(V_{21} , V_{41} , V_{14}), simulating stealthy attacks with obfuscated device communications Fig. 14. Motif dropout demonstrates changes in node density (V_{21} , V_{22} , V_{45}), simulating partial data loss from compromised nodes Fig. 15. This enables motif-based attention learning to prioritize rare motifs for anomaly detection, improving model robustness in IIoT environments.

The results in Table 6 show that motif augmentation boosts the APT detection approach. Pretraining with motif augmentation helps capture critical attack patterns missing in original motif hypergraphs. Techniques like temporal drift (+3.23% accuracy) and subgraph masking (+2.89% accuracy) improve model performance, especially in stage transition detection. Combining multiple augmentation techniques yields significant gains, achieving 98.71% accuracy and 88.70% stage transition accuracy.

Research Observation 3 How does MoGraM influence the convergence stability, adaptability, and detection accuracy of the Graph Convolutional Network (GCN) when handling evolving and obfuscated APT attack patterns?

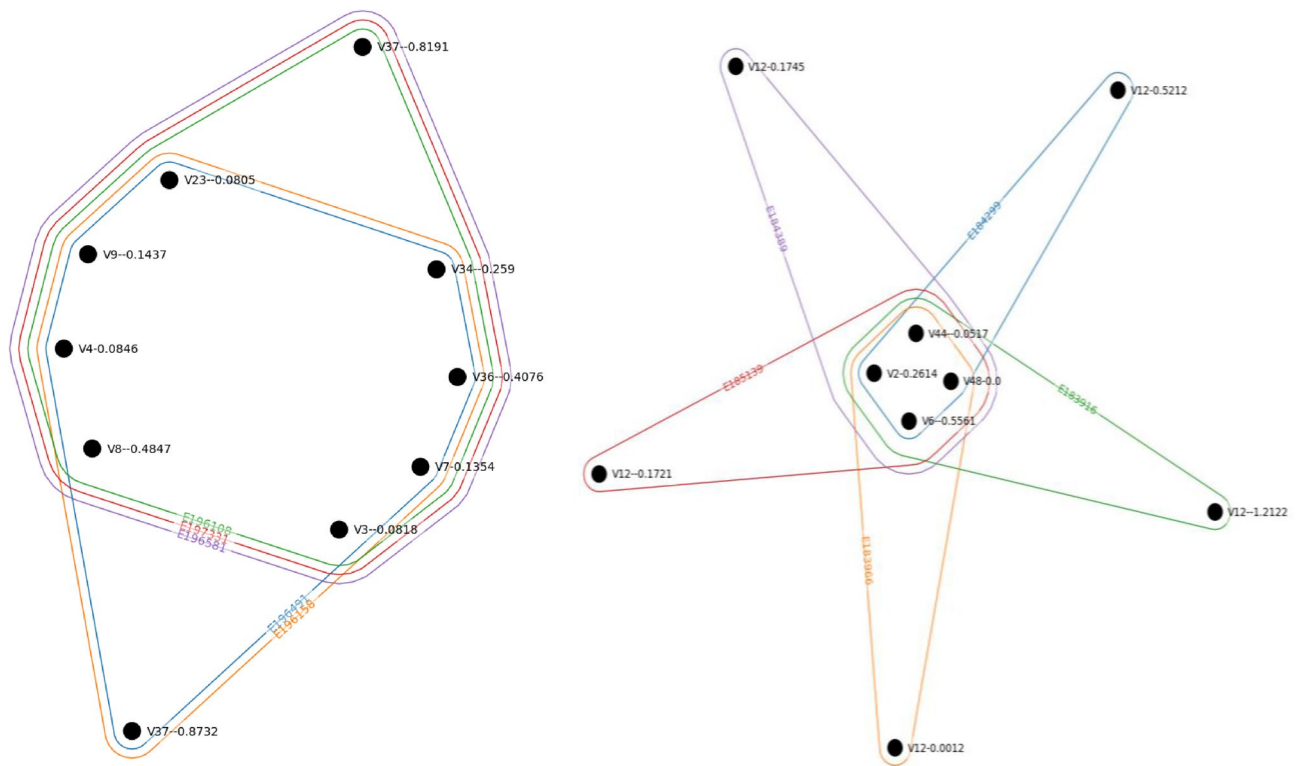


Fig. 13. Sample motif graphs occurred using the Motif Temporal Drift method.

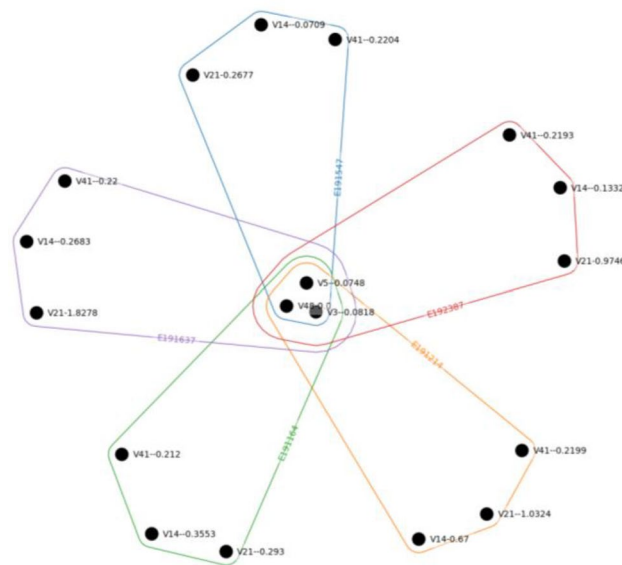


Fig. 14. Sample motif graphs occurred using the subgraph masking method.

The efficiency of the proposed MoGraM based GCN approach is validated using the comparison of stage wise metrics, comparing with exiting works, convergence stability, and adaptability for other datasets then ablation studies. The results confirms that the integration of MoGraM increase the performance of the GCN in APT attack detection, convergence and adaptability.

Table 7 presents the performance of the proposed SANGL approach using various metrics: accuracy, Precision, Recall, F1-Score, FPR, Graph Contrastiveness Score, Attention Divergence Score and Stage Transition Accuracy. The proposed approach shows strong performance with 98.71% accuracy, 98.43% precision, 98.31% recall, and 98.23% F1-score and 1.81% FPR. Of all the stage 6 exhibits, the lower FPR with 0.78% with accuracy of 98.64% and detection rate is 99.21% proves the robustness of the proposed approach. While stage transition

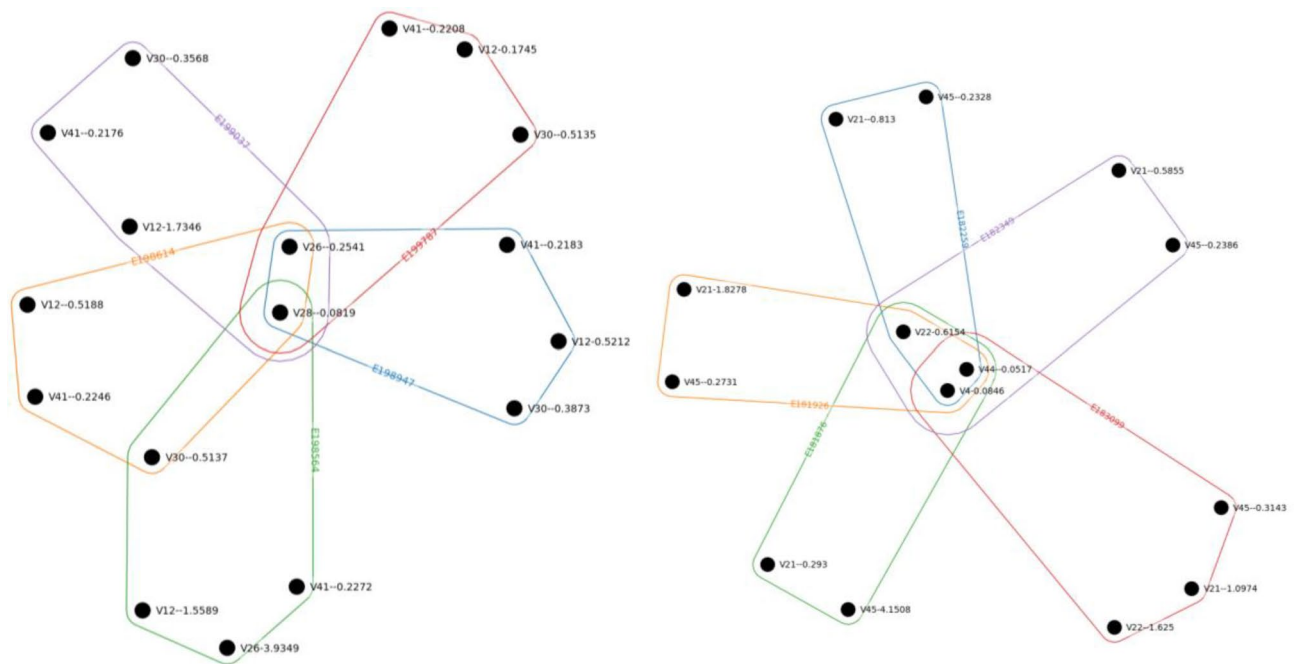


Fig. 15. Sample motif graphs occurred using the Motif Dropout method.

Augmentation Technique	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Stage Transition Accuracy (%)
No Augmentation	94.25	95.12	93.49	95.00	84.21
Motif temporal drift	97.48	95.79	95.78	96.36	86.74
Motif dropout	95.89	94.38	95.14	96.00	85.89
Subgraph masking	97.14	96.47	96.36	96.80	87.56
Combined motif augmentation process	98.71	98.43	98.31	98.23	88.70

Table 6. Comparing the performance of the motif augmentation process.

Labels	Performance Metrics (%)							
	Accuracy	Precision	Recall	F1-Score	FPR	Graph Contrastiveness Score	Attention Divergence Score	Stage Transition Accuracy
Stage 1	97.89	97.91	96.54	97.09	1.96	0.67	1.00	86.27
Stage 2	98.26	98.76	97.56	98.08	1.23	0.68	1.15	86.94
Stage 3	98.54	97.81	98.24	97.86	2.23	0.69	1.28	88.67
Stage 4	98.79	98.45	97.99	98.05	1.56	0.64	1.11	87.19
Stage 5	99.45	98.75	98.64	98.63	1.26	0.68	1.34	89.47
Stage 6	98.64	99.21	99.21	99.21	0.78	0.69	1.76	87.64
Stage 7	98.75	97.58	98.64	98.02	2.47	0.64	1.61	91.47
Stage 8	98.88	98.78	98.56	98.12	3.45	0.78	1.48	90.47
Benign	99.21	98.67	99.45	99.03	1.35	0.65	1.21	90.14
Overall Scores	98.71	98.43	98.31	98.23	1.81	0.68	1.32	88.70

Table 7. Comparison of the proposed SANGL approach in terms of various performance metrics.

accuracy is 88.70% shows that the SANGL approach temperately correlates the stage transitions of APT attack phases which often happens from stage 4 to stage 5.

Table 8 provides a detailed analysis conducted to validate the stability of the proposed approach. The validation is performed by considering four important hyperparameters: learning rate, dropout rate, motif embedding size, and motif weight factor. The proposed approach outperforms at 0.001 learning rate, 0.4 dropout rate, 0.5 weight of motif matrix and 64 motif embedding size. These values highlight the optimal configurations for the proposed SANGL approach. Also, it is noted that the performance remains consistent, with less fluctuations notably ± 0.01 in F1-Score, ensures the robustness and less sensitivity for parameter deviation.

Hyperparameters	Values	Performance Valuation					
		Accuracy	Precision	Recall	F1-Score	PR-AUC	Std. Dev (F1)
Learning rate	0.0005	98.43	98.21	98.06	98.05	98.12	±0.09
	0.001	98.71	98.43	98.31	98.23	98.26	±0.07
	0.005	97.96	97.62	97.46	97.58	97.78	±0.14
Dropout Rate	0.3	98.65	98.26	98.15	98.20	98.17	±0.06
	0.4	98.71	98.43	98.31	98.23	98.26	±0.07
	0.5	98.50	98.24	97.98	97.89	98.02	±0.08
Motif Embedding Size	32	98.14	97.98	97.56	97.85	97.94	±0.12
	64	98.71	98.43	98.31	98.23	98.26	±0.07
	128	98.49	98.34	98.21	98.17	98.01	±0.09
Weight Factor for Motif Matrix	0.3	98.22	97.98	97.84	97.85	97.45	±0.10
	0.5	98.71	98.43	98.31	98.23	98.26	±0.07
	0.7	98.36	97.94	98.14	98.10	98.12	±0.07

Table 8. Comparison of the SANGL learning parameters in terms of performance metrics.

Modules	Complexity	Explanation for notations
PyPCA	$(F^3 + NF^2)$	F^3 represents the Eigen decomposition over the feature covariance matrix, NF^2 the projection of all the observed samples in the reduced feature space
Motif Hypergraph Construction	$O(M * V^2)$	M represents the no. of motifs observed. V^2 is the pairwise motif subgraph representation.
Attention-based Motif Selection	$(**)$	Attention heads, no. of. Motifs and represents the feature dimensions of the motif embedding
MoGraM based GCN	$(***)$	represents iterations, represents the GCN layers, presents the no. of edges and is the hidden motif embedding dimensions
Time Complexity	$O(F^3 + NF^2 + MV^2 + KMF + ILEH)$	F^3 eigen decomposition, NF^2 feature space, MV^2 motif hypergraph representation, KMF attention based selection, $ILEH$ iterative GCN optimization using MoGraM
Space Complexity	$(+++)$	node matrix, edge embeddings and adjacency structures motif representations, attention based scoring

Table 9. Complexity analysis of the proposed SANGL Approach.

The computational complexity of the proposed approach is analysed with the time and space complexity of all the modules in the proposed approach. The following table [Table 9] explicitly explain the complexity for the proposed approach in terms of time and space and considered the complexity for each modules in the proposed SANGL approach.

Higher-order motif modelling enables SANGL to learn fine-grained stage transitions of APT attacks that are invisible to edge-only graphs. This gain in expressive power introduces additional overhead during motif enumeration and attention-based selection. However, by constraining motifs through top-k motifs pruning and exploiting sparse IIoT graphs, SANGL remains computationally feasible for large-scale APT detection while offering substantially improved relational modelling.

To ensure the convergence stability, Fig. 16 presents the graphs to highlight the accuracy of the proposed approach vs. epoch and loss vs. epoch. It is clearly captured that the training accuracy slightly converged around 92.4% to 98.7% and testing accuracy converged earlier around 85.4% shows that the proposed approach is working well for unseen data. While in the case of the loss graph, the testing loss converges earlier at 0.2 in 25 epochs, whereas the training loss converges around 0.1 in 50 epochs. Hence, the SANGL approach is working well for both training and testing data and has the potential for a good fit with good generalizability.

Table 10 compares various graph-based and deep learning models for APT detection process. Three performance metrics are considered for comparison: Accuracy, Precision and recall. GCN²⁹ model is presented with the accuracy of 90.51% used synthetically generated malware dataset which is 8% lesser than the proposed SANGL approach. Similarly, GCN⁵⁰ article presents the accuracy with 95.90% which is 3% lesser than proposed approach^{51,52,54,57,58} used graph and deep learning based approaches and validated the model with the simulated IIoT related datasets and described with the results 96.97%, 97.76%, 99.89%, 97.22%, 97.95%. Among these models, MAD v1 (MLP) & MAD v2 (MAML)⁵⁴ model presents higher accuracy than the proposed approach but it fails to address the higher detection rate. As proposed SANGL approach has higher accuracy along with detection rate which shows the robustness of the model. While comparing the proposed approach with other recent hypergraph based approaches [44 & 60] SANGL outperforms with 1.56% & 8.68% increase in accuracy respectively. NB, DT, RF, SVM, LR, Bagging, XGBoost and AdaBoost⁵⁶ models for APT detection by implementing in the NSL KDD dataset with the stages of APT. Though it result in 99% and 100% the model

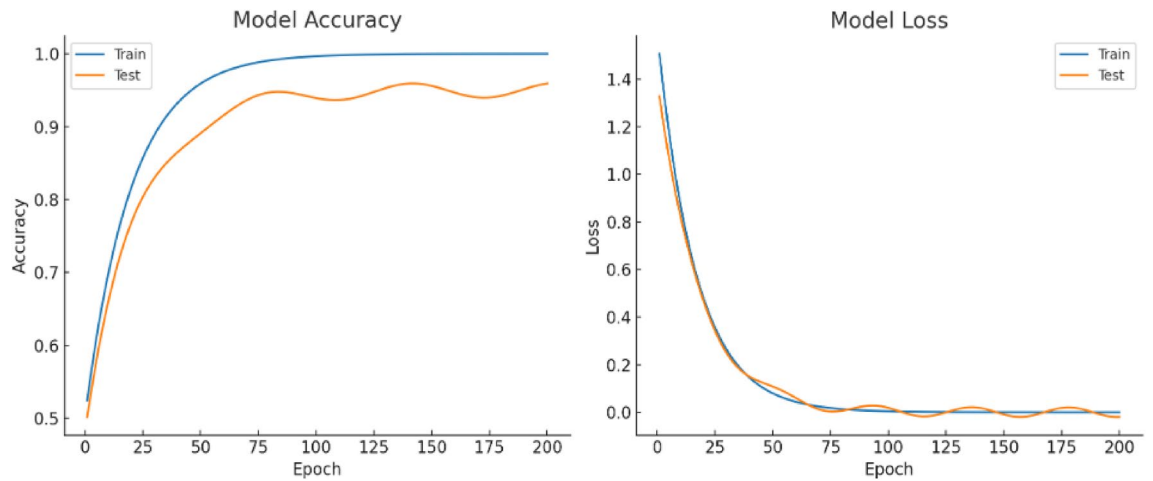


Fig. 16. Accuracy and loss graphs of proposed SANGL Approach.

Model	Dataset used	Performance Metrics (%)		
		Accuracy	Precision	Detection Rate
GCN ²⁹	Synthetic malware data	90.57	90.51	90.57
GCN ⁵⁰	Synthetic malware data	95.90	95.10	95.10
Graph Attention Network ⁵¹	DAPT20, Edge I-IoT	96.97	95.72	96.58
Graph Attention Network & GraphSAGE ⁵²	Cyber and Physical domain dataset	97.76	88.46	90.20
Graph Neural Diffusion Network ⁵³	Linux kernel-based APT audit logs	-	96.03	-
MAD v1 (MLP) MAD v2 (MAML) ⁵⁴	CICAPT-IIoT	99.89	99.10	91.10
Bi-LSTM + Attention + DGCNN ⁵⁵	Malware Capture CTU-13	99.10	92.21	91.16
HGCNN ⁴⁴	Real world dataset	97.15	-	-
NB, DT, RF, SVM, LR, Bagging, XGBoost and AdaBoost ⁵⁶	NSLKDD	99.90	99.90	100.00
Hybrid Multistage Intrusion Detection System ⁵⁷	CICToN IoT	97.22	97.22	97.22
	CIC IDS 2017	97.22	97.22	97.22
	NSL KDD	99.92	99.92	99.92
	KDD Cup 99	97.22	97.72	97.92
LSTM Autoencoder ⁵⁸	Industrial control system traffic data set (GP)	97.95	98.01	96.63
	UNSW-NB15	97.62	97.69	96.01
High Discrimination APT Intrusion Detection System ⁵⁹	NSLKDD dataset	81.43	-	-
Hypergraph Attention Network ⁶⁰	Unicorn	90.03	-	91.15
SANGL (Proposed Approach)	CICAPT- IIoT	98.71	98.39	98.28

Table 10. Comparison of the proposed SANGL approach with previous works.

Models	Performance Metrics (%)				
	Accuracy	Precision	Recall	F1-Score	FPR
GCN	88.21	91.26	91.01	90.49	6.74
PyPCA_GCN	89.78	90.28	91.53	92.16	3.59
PCA_GCN	87.59	89.21	89.88	91.22	4.21
PCA_MoGraM GCN	92.49	95.42	93.76	92.39	3.26
MoGraM-based GCN	91.47	89.58	89.98	89.99	3.67
Motif graph-based GCN	91.45	94.12	89.27	92.14	3.78
PyPCA Motif-based GCN	94.47	93.21	95.48	96.12	4.21
PyPCA-Motif graph-based MoGraM -GCN	98.71	98.39	98.28	98.26	1.62

Table 11. Ablation study of the proposed SANGL approach.

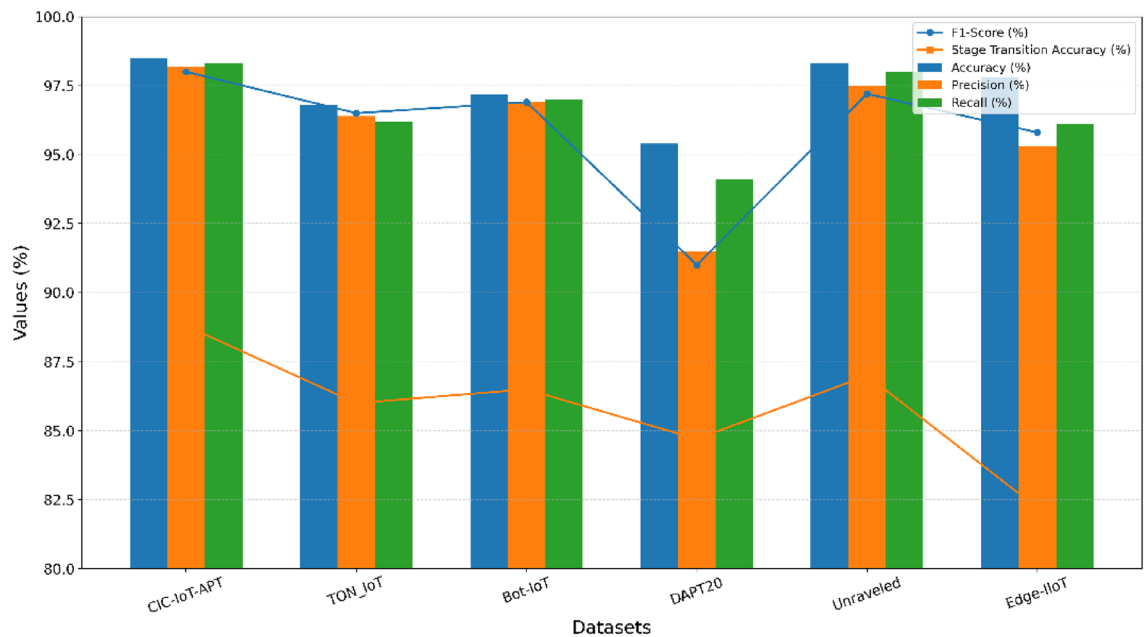


Fig. 17. Comparison of various datasets using performance metrics.

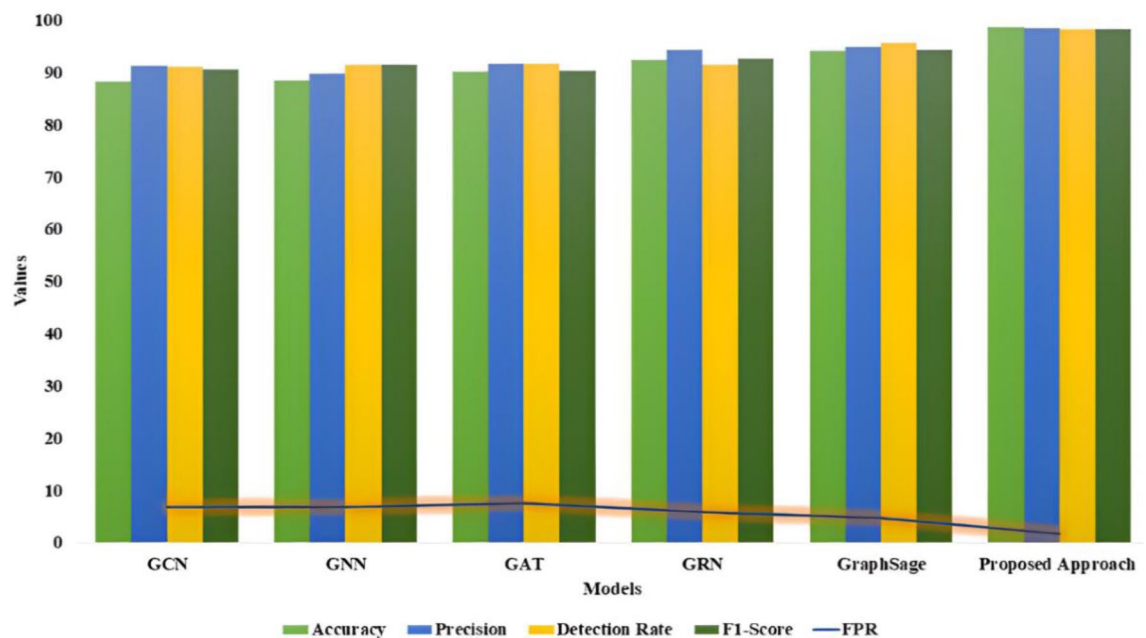


Fig. 18. Comparison of the proposed approach with various baseline models.

lacks in addressing the overfitting issues, reliability and robustness. As a result, it is observed that our proposed approach outperforms all the existing works in various aspects.

Table 11 presents the performance metrics compared across various models to showcase the importance of integrating the MoGraM for GCN. While integrating the accuracy score is 98.71% which is 4.24% higher than the PyPCA Motif based GCN and 7.24% higher than the MoGraM GCN model. Also, 6.22% greater when compared with PCA_MoGraM GCN model. Likewise the integration of MoGraM method in the proposed approach outperforms all the other models in all such performance metrics. Especially in detection rate the proposed SANGL approach is 8.3% better than the MoGraM GCN model. This shows the effectiveness of MoGraM method for GCN.

SANGL approach demonstrates excellent performance across various datasets Fig. 17, with the CICAPT-IIoT dataset showcasing the best results: accuracy (98.71%), precision (98.39%), detection rate (98.28%), F1-

score (98.26%), and stage transition accuracy (88.47%). This outperforms other datasets, including DAPT20, Unravalled, Edge-IIoT, TON_IoT and BoT-IoT, indicating that the proposed approach is particularly effective for the CICAPT-IIoT dataset, likely due to its suitability for the dataset's characteristics and patterns.

SANGL approach outperforms various baseline methods, including GCN, GAT, GRN, GNN, and GraphSage, in detecting anomalies, as evident from the performance metrics Fig. 18 With an accuracy of 98.71%, a precision of 98.39%, a detection rate of 98.28%, and an F1-score of 98.26%, the proposed approach demonstrates exceptional performance. Moreover, it achieves a remarkably low False Positive Rate (FPR) of 1.62%, significantly outperforming the baseline models, which have FPRs ranging from 4.58% to 7.45%. This suggests that the proposed approach is not only highly accurate but also robust in minimizing false alarms, making it a superior choice for anomaly detection tasks.

To enhance the generalizability of the proposed approach, significance test are conducted. Primarily, to determine the appropriate significance test, the Shapiro-Wilk Normality test Equation (28) was implemented to test the differences between the proposed SANGL approach and the other baseline models across all APT attack stages. The default value for the test is defined with $p = 0.05$. If the test is predicted with the value $p > 0.05$ then the difference is absolutely normal hence paired t-test is performed. If $p < 0.05$ then the differences are not normal hence Wilcoxon signed rank test is performed.

$$ShapiroWilkNormalitytest = ProposedApproach_i - Baselineapproach_i \tag{28}$$

Where i defines the performance metrics of the approaches.

Table 12 presents the significance test values for the proposed approach compared to the baseline approaches. The statistical analysis involved Shapiro–Wilk normality tests to evaluate the paired stage-wise differences between SANGL and each baseline model across five key metrics: Accuracy, Precision, Recall and F1-score. The differences versus GCN followed an approximately normal distribution ($p > 0.05$), warranting the use of paired t-tests. In contrast, differences versus GNN and LSTM showed slight deviations from normality ($p \leq 0.05$), necessitating the application of Wilcoxon signed-rank tests. The results revealed that SANGL significantly outperformed all baseline models across all metrics ($p < 0.01$), confirming that the observed performance gains are statistically meaningful rather than mere random variation. Notably, the largest improvements were observed in F1-score and Recall, underscoring SANGL's superior ability to model complex APT stage transitions. Additionally, the reduced FPR highlights the model's enhanced robustness and fewer false alarms, demonstrating its effectiveness in accurately detecting APT attacks.

Figure 19 highlights the effectiveness of the proposed techniques, highlighting their impact on overall performance. An ablation study was conducted to validate the approach, demonstrating its superiority over individual components. The results confirm that the integrated approach yields significant improvements, underscoring the value of combining these techniques.

Limitations of the SANGL approach

Although the proposed SANGL approach exhibits determinants for the APT detection process in IIoT environments, certain limitations remain to be addressed in future work. The limitations includes, Dataset diversity, class imbalance, temporal and sequential dependencies, concept drift and adaptability. The proposed approach outperformed other baseline models for both the simulated and benchmark IIoT datasets, achieving higher accuracy and stage transition accuracy. However, the observed dataset doesn't provide the higher variety and randomness for real-world APT attacks. From the experimental results, it is evident that the per-class stage accuracy remains lower for stages such as command & control and persistence. Though imbalance issues are addressed, these stages presented fewer samples which causes the model to deteriorate its performance. The proposed Motif graph captured the structural and motif-level representations; however, it failed to present the temporal-level embeddings. As APT tactics evolve, the underlying APT data will get distributed over time, integrating with the adaptive online learning to maintain the durability of the dynamic environments.

Models	Performance Metrics	Shapiro test Values (%)	Test Used	Test Values (%)
GCN	Accuracy	0.67	Paired t-test	0.0003
	Precision	0.61	Paired t-test	0.0005
	Recall	0.72	Paired t-test	0.0007
	F1-Score	0.69	Paired t-test	0.0004
GNN	Accuracy	0.59	Paired t-test	0.0002
	Precision	0.04	Wilcoxon test	0.0012
	Recall	0.03	Wilcoxon test	0.0010
	F1-Score	0.06	Paired t-test	0.0011
LSTM	Accuracy	0.01	Wilcoxon test	0.0001
	Precision	0.02	Wilcoxon test	0.0002
	Recall	0.01	Wilcoxon test	0.0001
	F1-Score	0.03	Wilcoxon test	0.0001

Table 12. Significance test for proposed SANGL approach vs. baseline approaches.

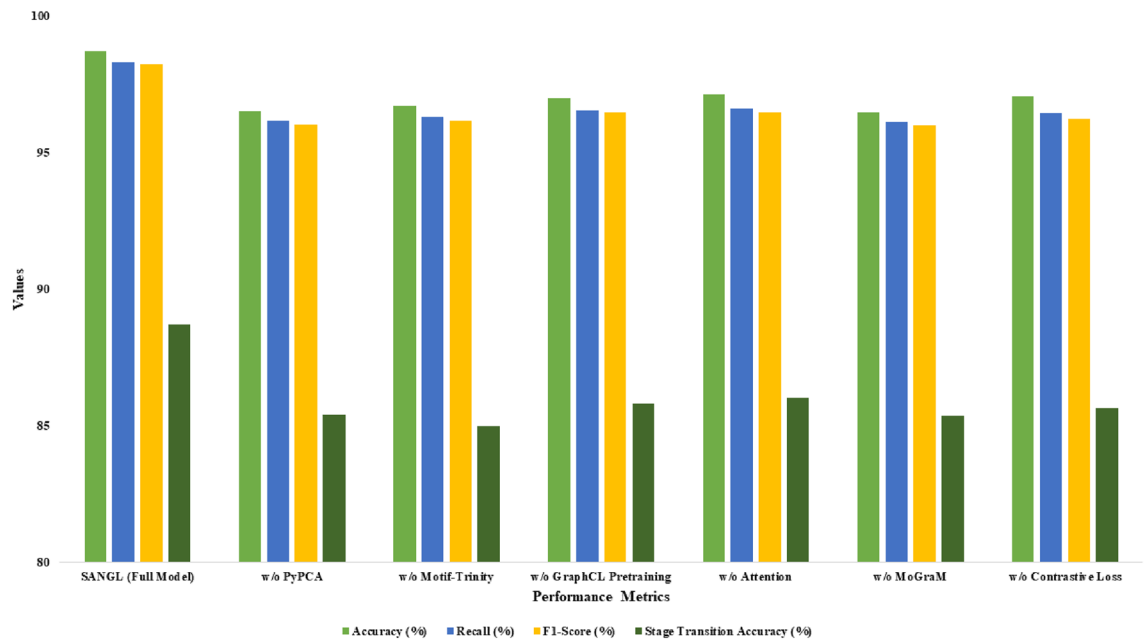


Fig. 19. An ablation study of the proposed approach with various performance Metrics.

Conclusion and future work

APT attacks evolve in all network environments, specifically in the IIoT environment, which necessitates exertion in the detection process. The limitations include a high false positive rate, being computationally inefficient and difficult in correlating the stages of APT attack. Considering this, a novel Sequential Analysis of Network patterns using graph-based Learning approach (SANGL) is proposed. This approach integrated the PyPCA method, motif hypergraph method and optimized GCN method. The proposed approach effectively attributed to extract critical attack patterns and refined features through motif-based clustering, enabling a comprehensive understanding of APT attack scenarios. Pretraining mechanism and Multi-objective Gradient Modulator (MoGraM) based graph convolution network further enhanced the model's performance, captured the evolving APT patterns and stage transitions. SANGL had demonstrated exceptional performance in the detection of APT attack transition with an overall accuracy of 98.71%, precision of 98.39%, recall of 98.28%, and F1-score of 98.26%, and maintained a low False Positive Rate (FPR) of 1.62%. Moreover, the proposed approach is evaluated using two graph metrics such as Stage Contrastiveness Score of 0.68 and a Stage Transition Accuracy of 88.7% demonstrating its ability to capture stage-wise dependencies and temporal transitions. These metrics highlighted the model's robustness, scalability, and adaptability in handling complex, multi-stage APT behaviours. From these resilient performances and potential for future enhancements, the proposed SANGL approach is efficient in the field of IIoT security. Although the model performs well, it has some limitations while addressing issues such as dataset bias and scalability. The attackers of simulated dataset does not include all the tactics of real-time attack hence, training the model on various simulated datasets may be less effective for multi-stage attack detection in terms of false positives and false negatives. While in the case of scalability, if the model is deployed in a real-time environment, it leads to more computation, which reduces the time of precise prediction. Concerning this, future research directions will focus on expanding dataset diversity, enhancing interpretability, exploring real-time deployment, and assessing edge deployment feasibility in IIoT environments, considering energy and resource constraints to improve the model's capability and practical applicability in securing industrial infrastructure against evolving APT threats.

Data availability

The datasets analysed during the current study available from the corresponding author on reasonable request.

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Declarations

Competing interests

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